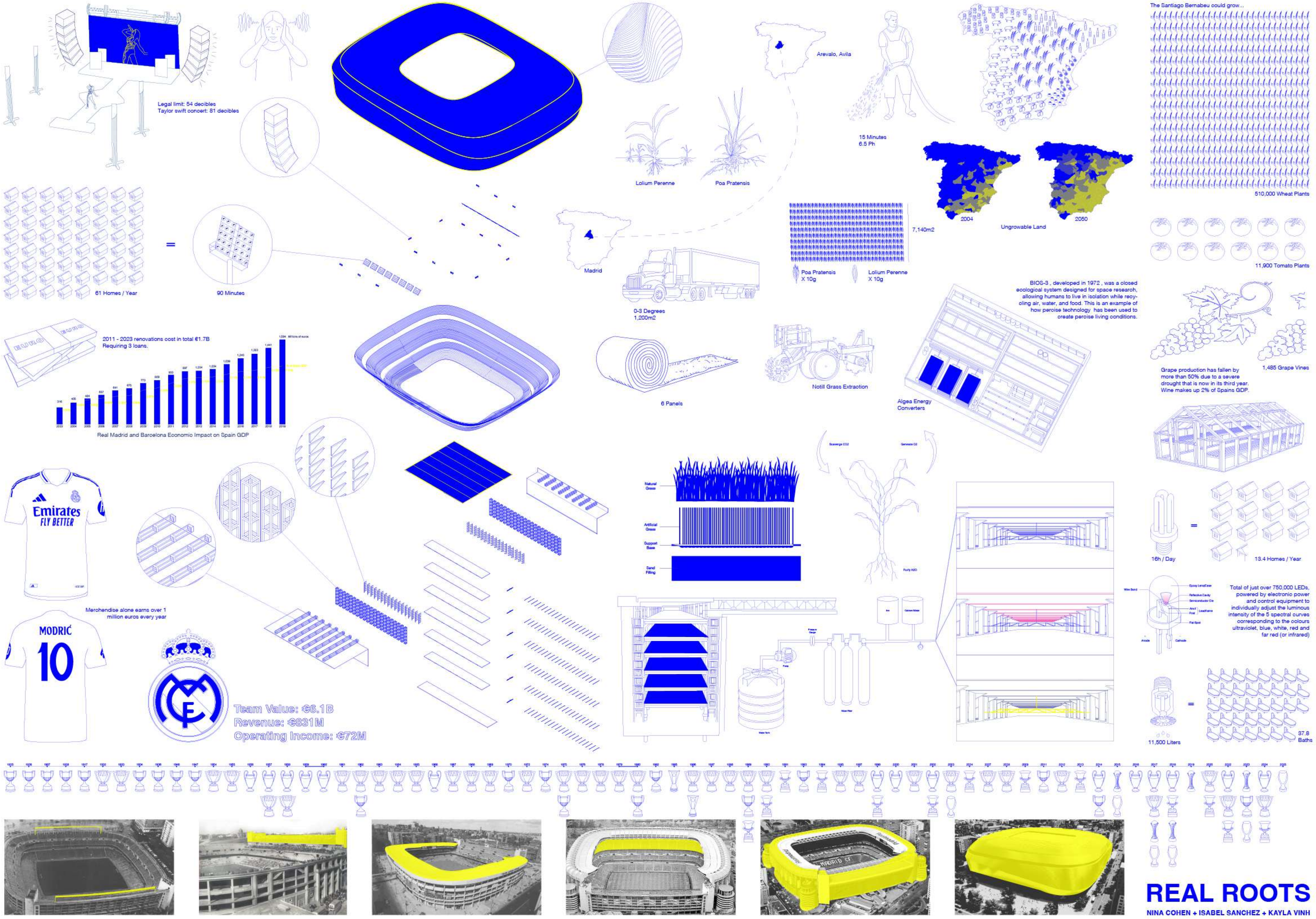
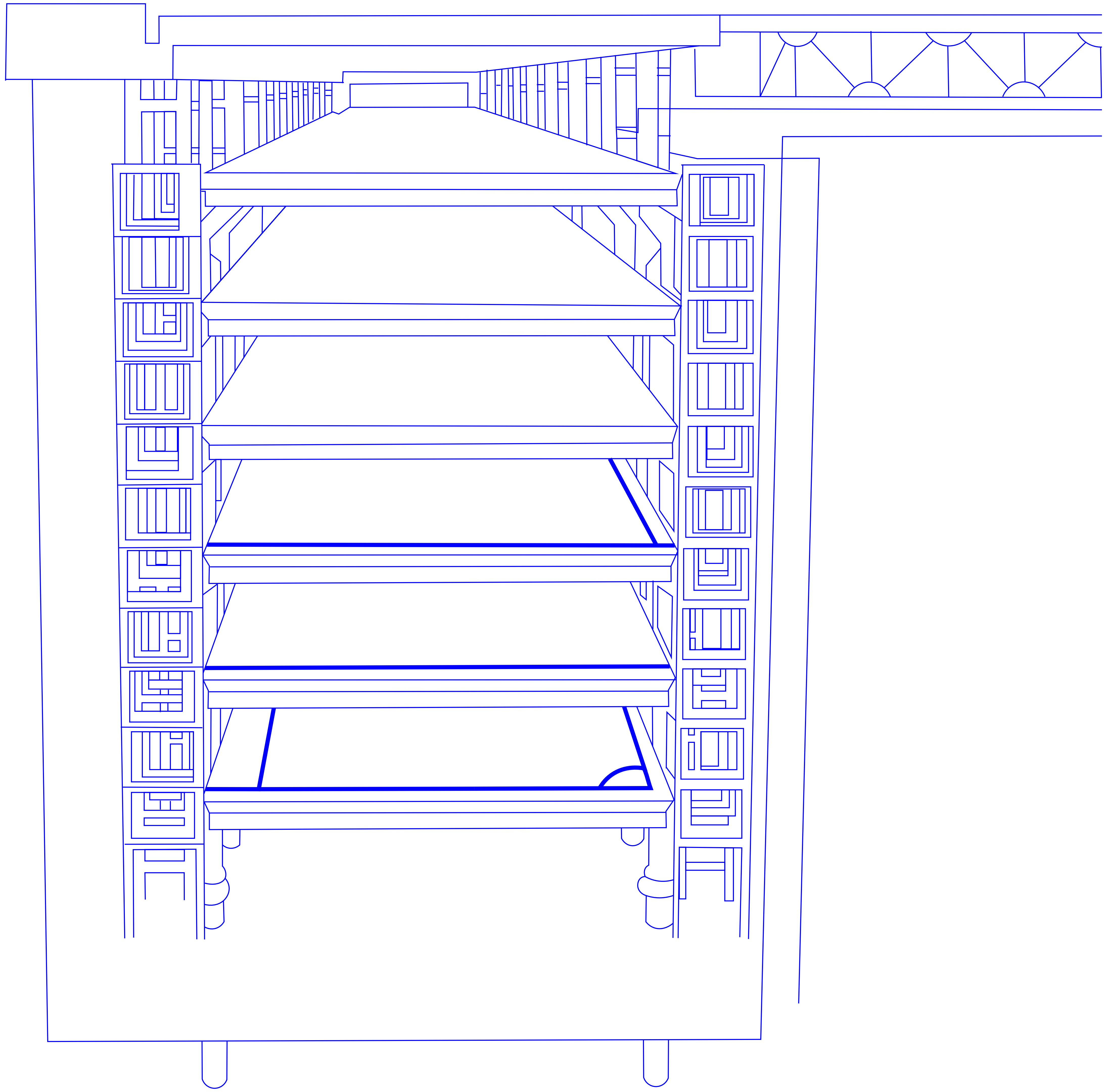
REAL ROOTS

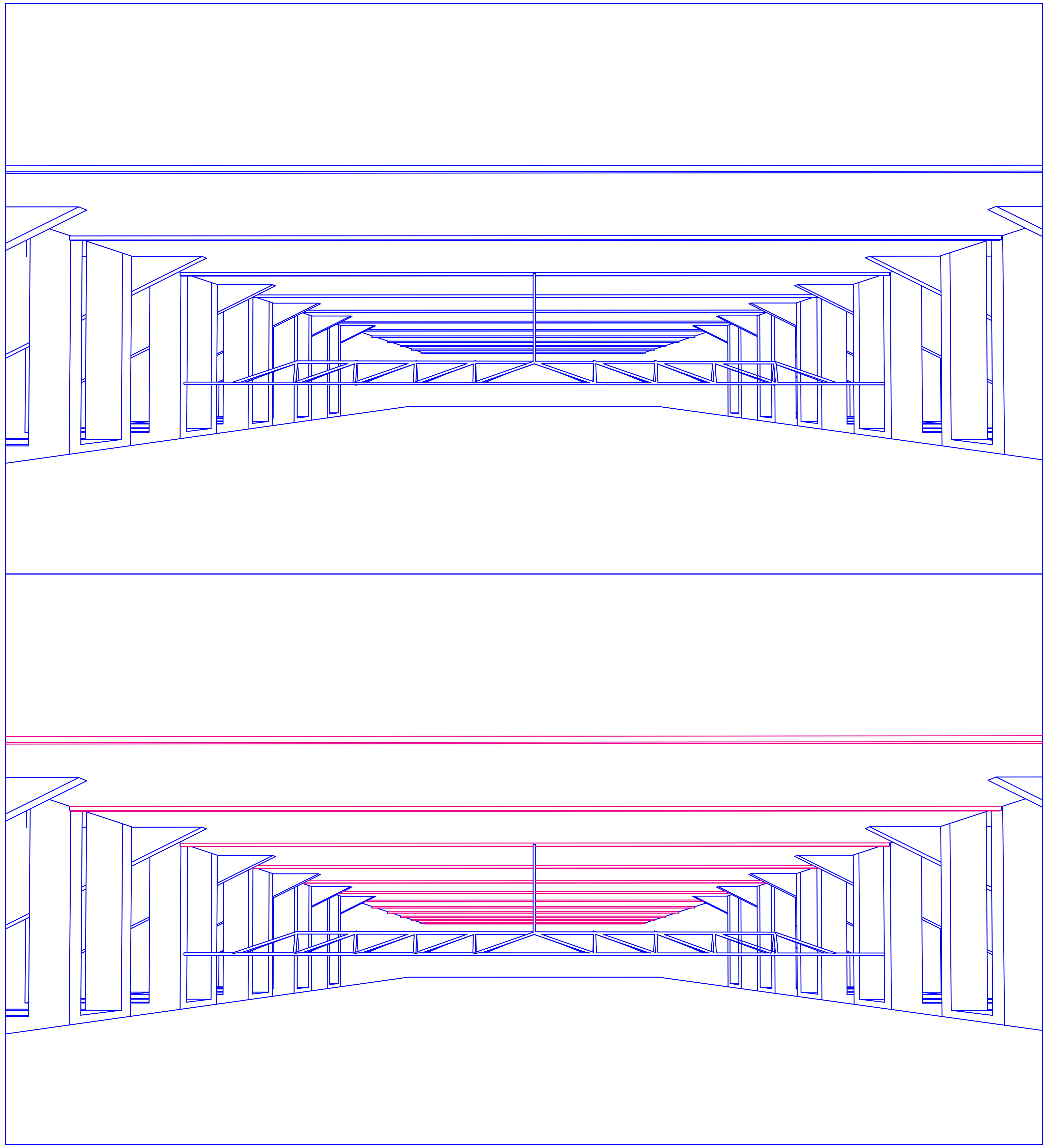
Design Studio III

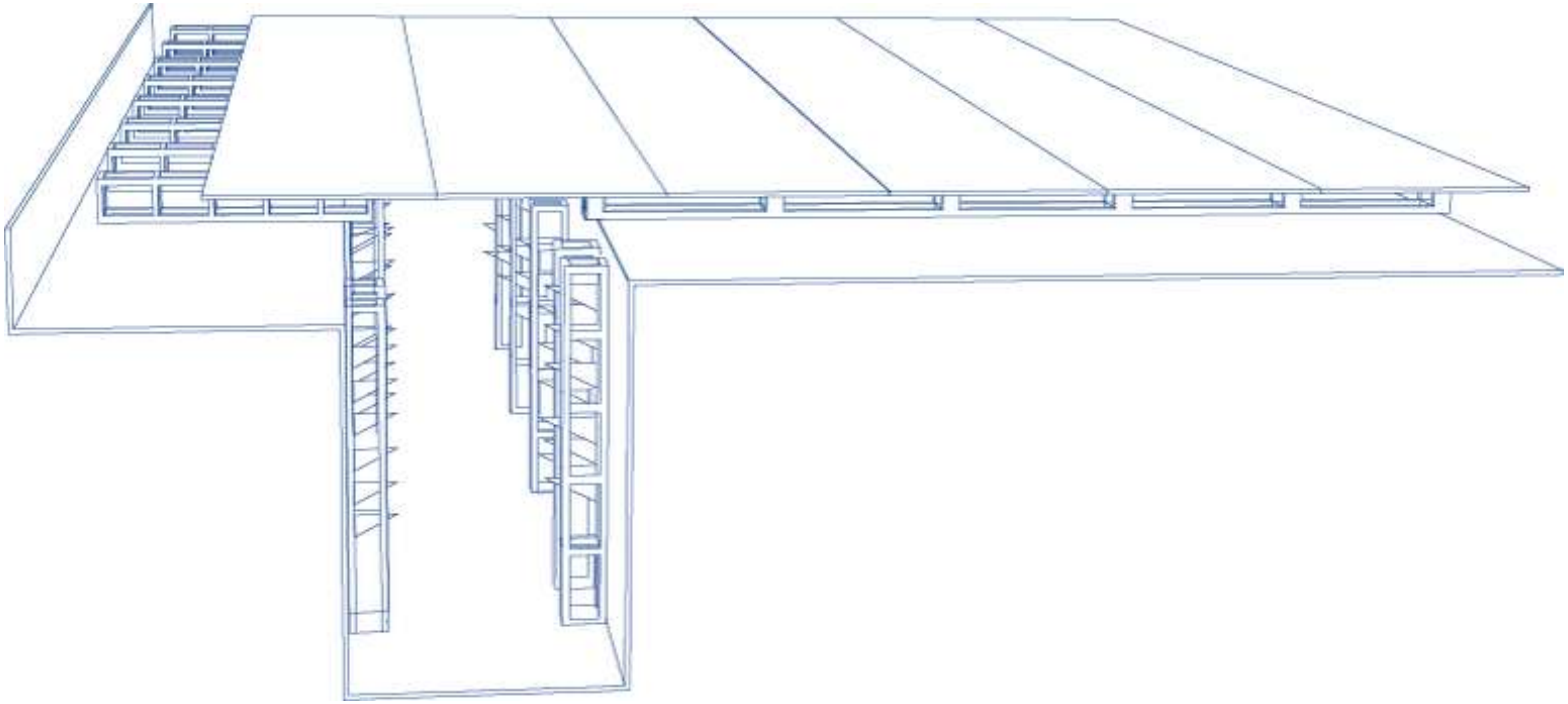
Nina Cohen, Isabel Sanchez, Kayla Vinh

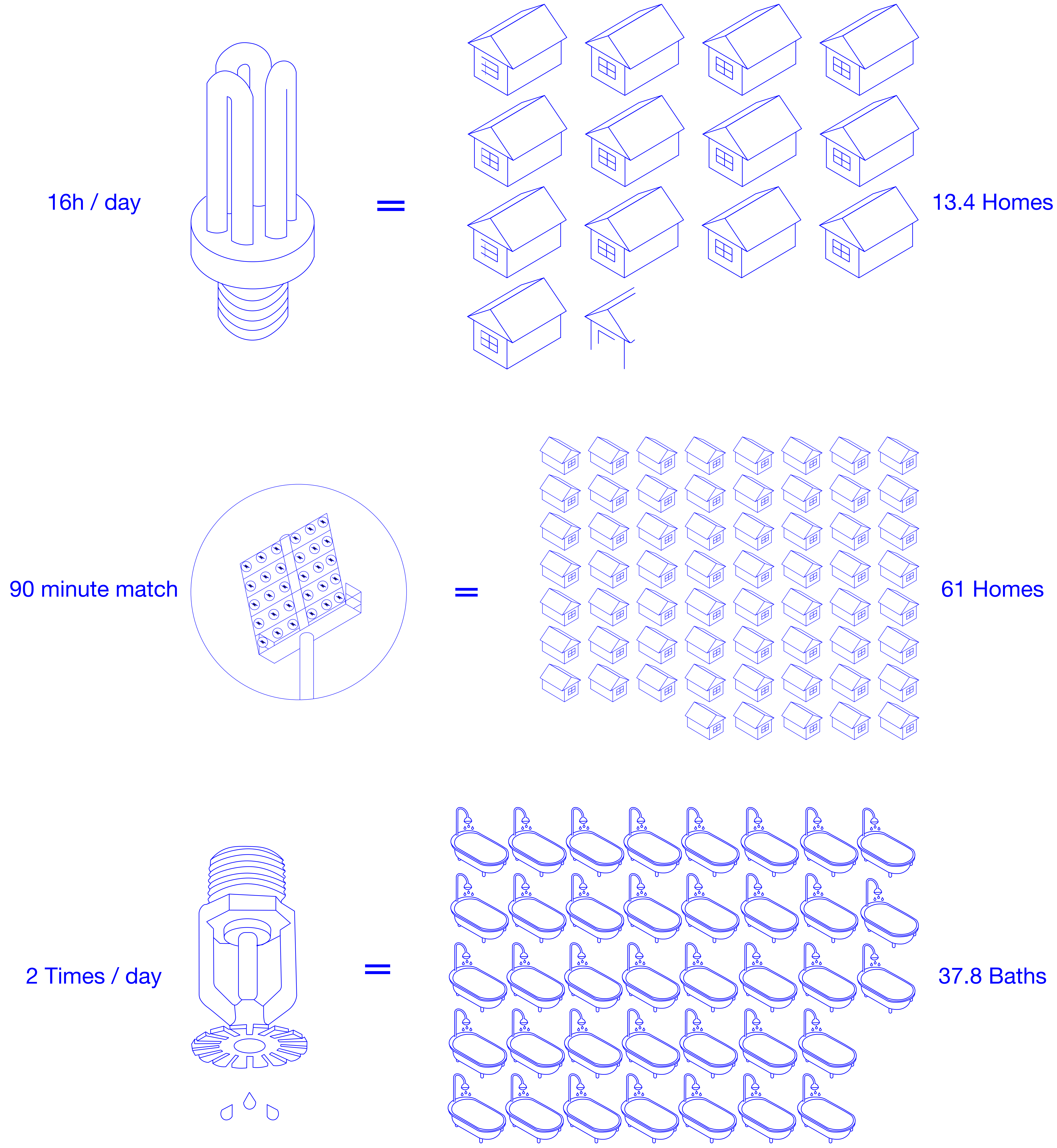


Legal limit: 54 decibles
Taylor swift concert: 81 decibles
61 Homes / Year
=
90 Minutes
2011 - 2023 renovations cost in total €1.7B
Requiring 3 loans.
Real Madrid and Barcelona Economic Impact on Spain GDP
Emirates
FLY BETTER
MODRIĆ
10
Merchendise alone earns over 1 million euros every year
Team Value: €6.1B
Revenue: €831M
Operating Income: €72M
Lolium Perenne
Poa Pratensis
Madrid
Arevalo, Avila
15 Minutes
6.5 Ph
0-3 Degrees
1,200m2
6 Panels
Notill Grass Extraction
7,140m2
Poa Pratensis
X 10g
Lolium Perenne
X 10g
2004
Ungrowable Land
2050
BIOS-3 , developed in 1972 , was a closed ecological system designed for space research, allowing humans to live in isolation while recycling air, water, and food. This is an example of how percise technology has been used to create percise living conditions.
Algea Energy Converters
Natural Grass
Artificial Grass
Support Base
Sand Filling
Scavenge CO2
Generate O2
Purify H2O
The Santiago Bernabeu could grow...
510,000 Wheat Plants
11,900 Tomato Plants
1,485 Grape Vines
Grape production has fallen by more than 50% due to a severe drought that is now in its third year. Wine makes up 2% of Spains GDP.
16h / Day
=
13.4 Homes / Year
Wire Bond
Epoxy Lens/Case
Reflective Cavity
Semiconductor Die
Anvil
Post
Leadframe
Flat Spot
Anode
Cathode
Total of just over 750,000 LEDs, powered by electronic power and control equipment to individually adjust the luminous intensity of the 5 spectral curves corresponding to the colours ultraviolet, blue, white, red and far red (or infrared)
11,500 Liters
=
37.8 Baths
REAL ROOTS
NINA COHEN + ISABEL SANCHEZ + KAYLA VINH



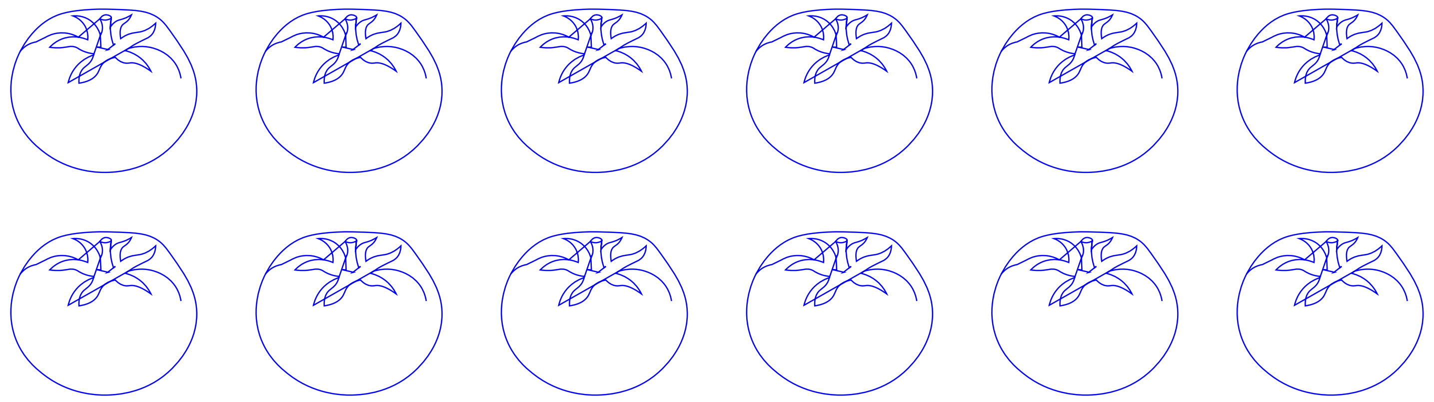




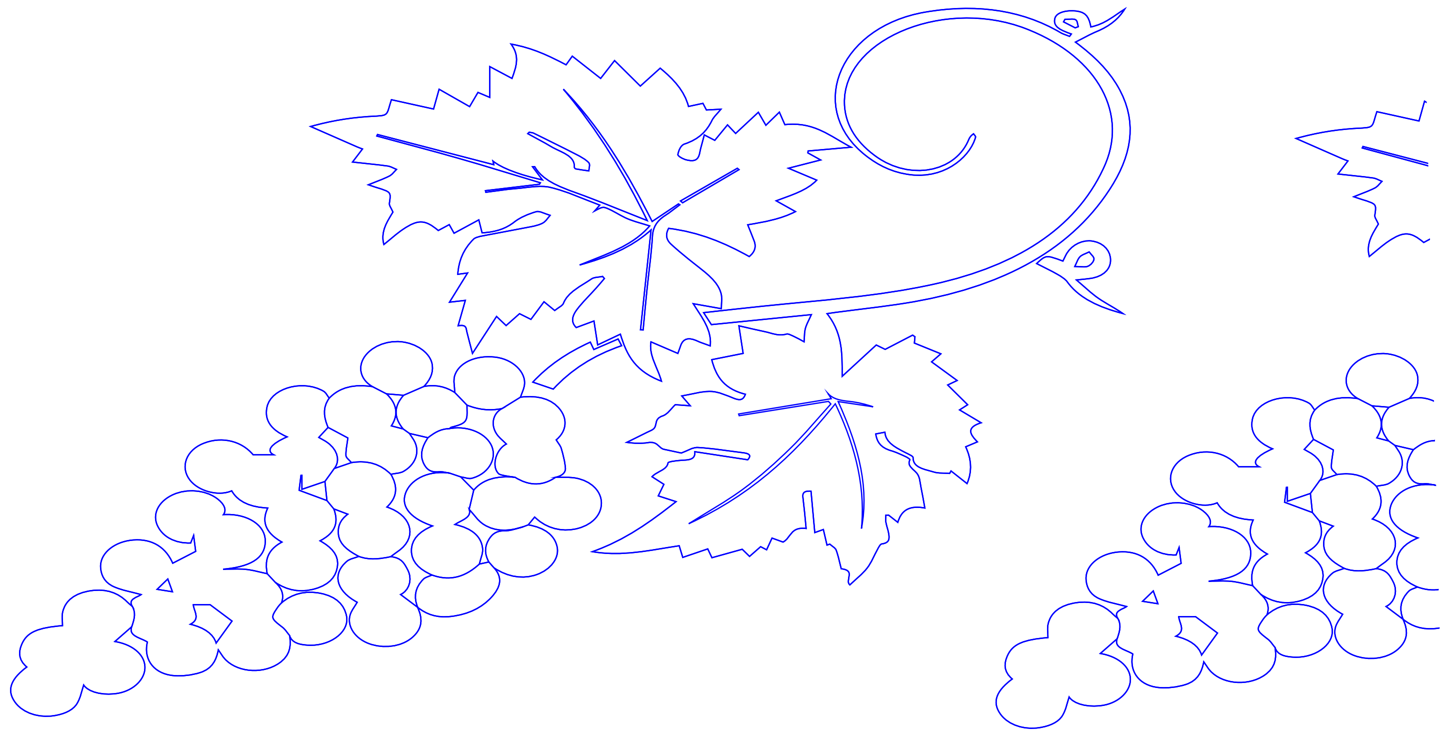

16h / day
13.4 Homes
90 minute match
61 Homes
2 Times / day
37.8 Baths

The Santiago Bernabeu could grow...

510,000 Wheat Plants



11,900 Tomato Plants



1,485 Grape Vines

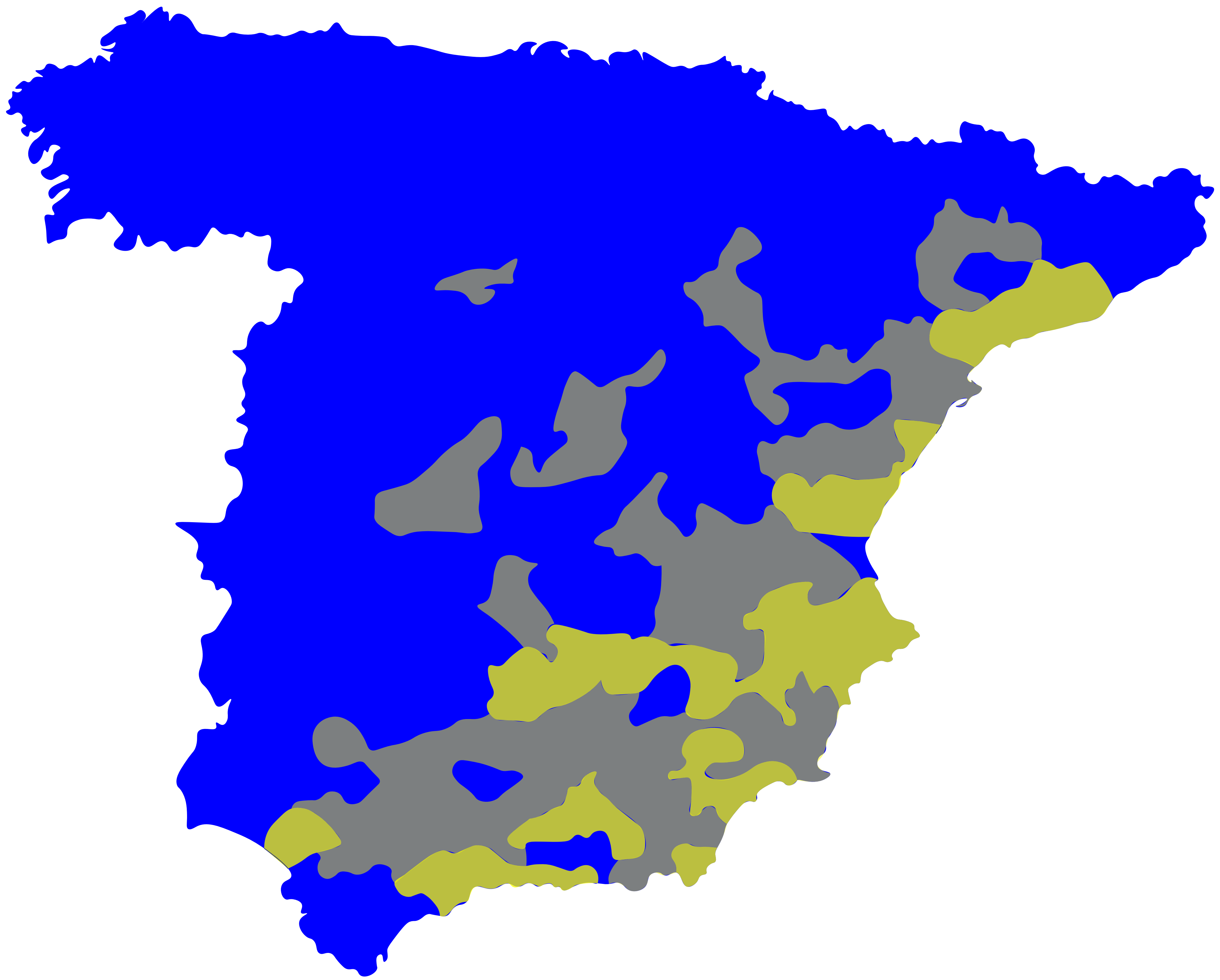
THE FUTURE

2023, 26% of Spain’s territory was in a situation of prolonged drought.

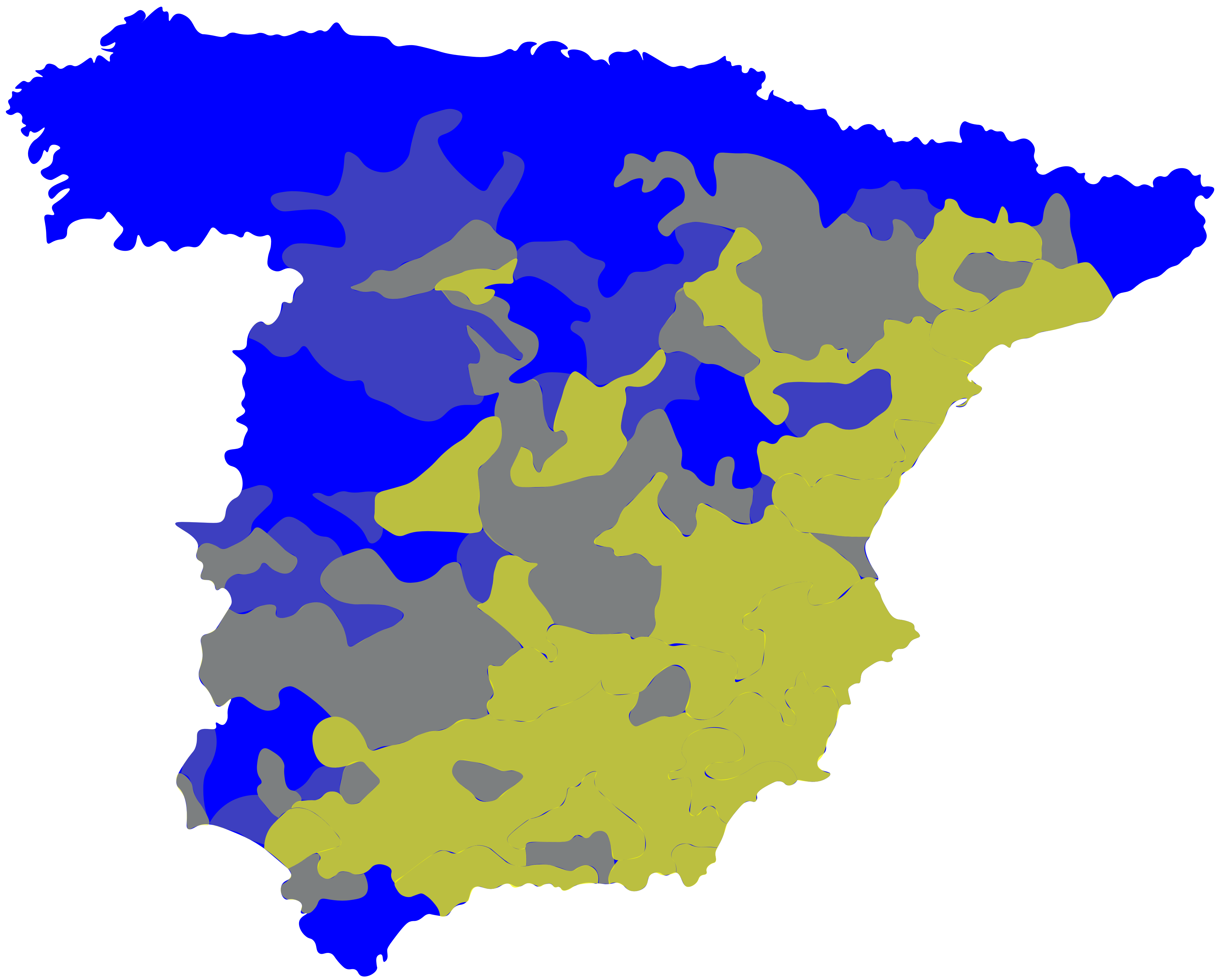
Between 2030 and 2050 Spain will loose €375 billion due to drought.

Spain maize production to decrease 80% by 2050.

2050, the global population will be 10 billion people, requiring a 60% increase in food.



2004



2050

Spain Consumption Per Capita

Wheat = 100.9 (kg)

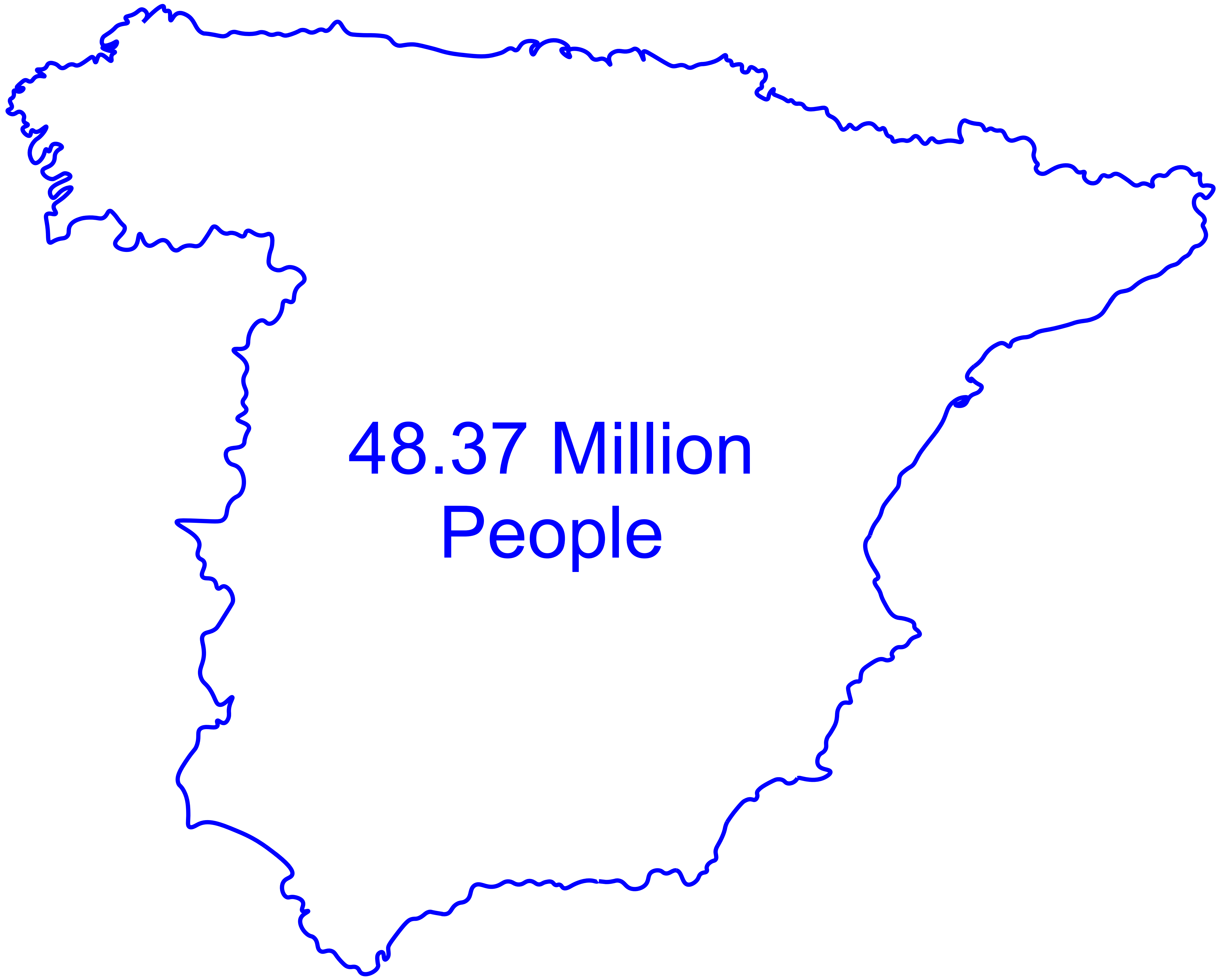
Potato = 60 (kg)

Tomato = 13.5 (kg)

Onions = 7.1 (kg)

Rice = 6 (kg)

Peppers = 5.1 (kg)



Santiago Bernabéu is USING advanced TECHNOLOGY to further consumerism in THE ENTERTAINMENT INDUSTRY wasting precious resources we will need for THE FUTURE.

By integrating food production, resource conservation, and community engagement, the advanced technology used by the Santiago Bernabéu will be reimagined to offer scalable solutions to address food scarcity in Madrid's future.

HOMEGROWN

The perfect home for you and your family in the case of severe famine :)

Escalating resource scarcity and the mounting challenges of climate change demand innovative approaches to sustainable agriculture. By reimagining the Santiago Bernabéu Stadium's hypogeum, it will be repurposed as a model for sustainable agricultural solutions across multiple scales. By **integrating food production with space optimization and community engagement**, future food demands as well as resource conservation will be addressed.

At the home level, the hypogeum concept is **adapted to enable households to cultivate essential crops** like tomatoes, potatoes, peppers, and onions in energy-efficient environments. This approach enhances food accessibility and empowers families to achieve self-sufficiency through year-round production of fresh produce, tailored to individual needs and capabilities.

At the entertainment scale, the hypogeum transforms the Santiago Bernabéu into a **destination for agritourism and environmental appreciation**. Seasonal, plant-based attractions such as strawberry picking, lavender strolls, and sunflower fields create immersive and interactive agricultural experiences. By cultivating non-native species year-round, this model merges sustainability with novel, visitor-focused experiences that educate and inspire environmental stewardship.

On an urban scale, the hypogeum becomes a city-sized agricultural system, designed to **produce staple crops** like wheat and rice. By reducing dependency on long-distance food transport and imports, this model alleviates the strain on global supply chains and promotes local food security. This sustainable solution addresses metropolitan food demands while minimising environmental impacts and maximising land-use efficiency.

Casa de las Flores

Secundino Zuazo

1931

Chamberí, Argüelles

Rationalist approach

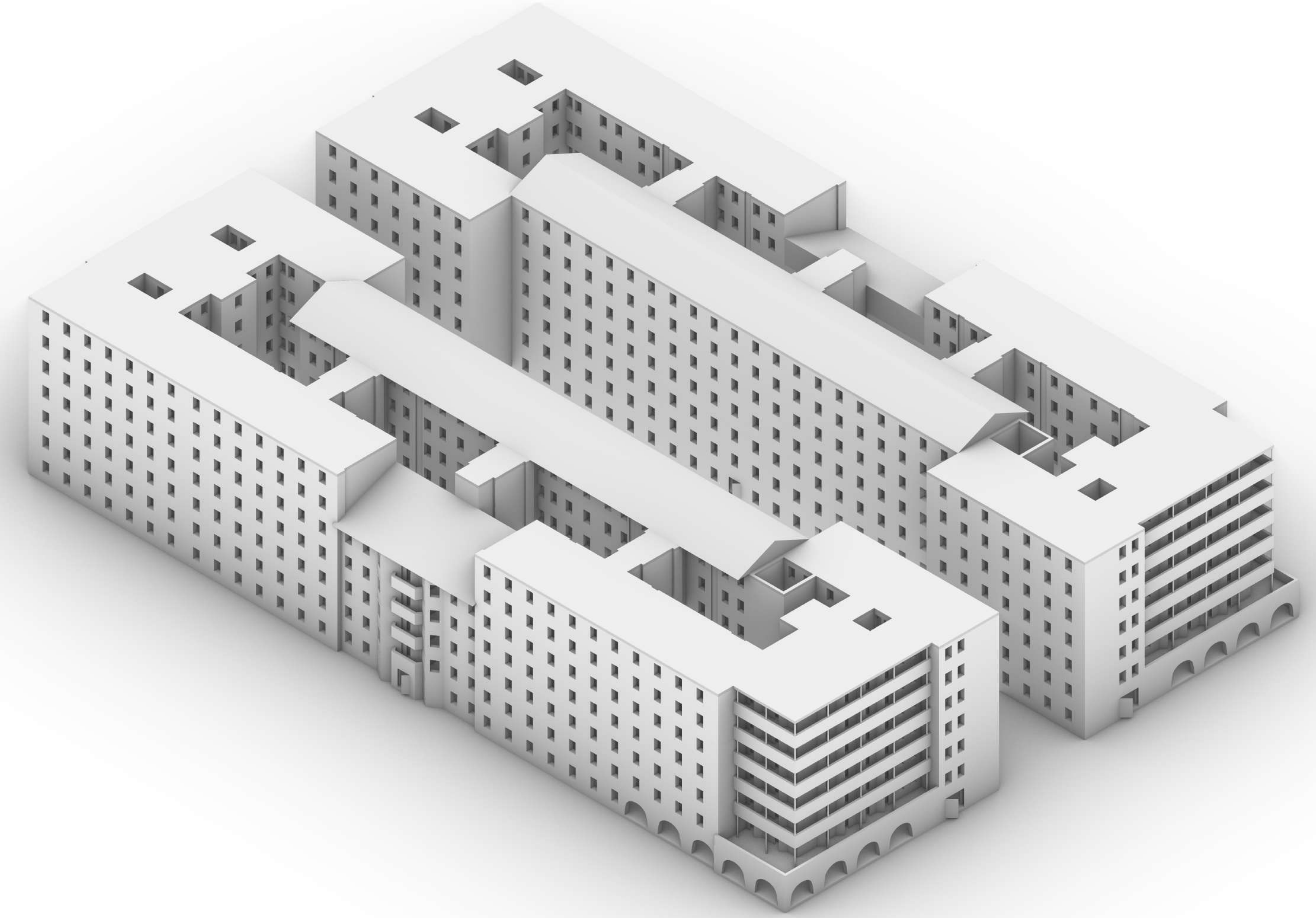
Casa de las Flores

Secundino Zuazo

1931

Chamberí, Argüelles

Rationalist approach



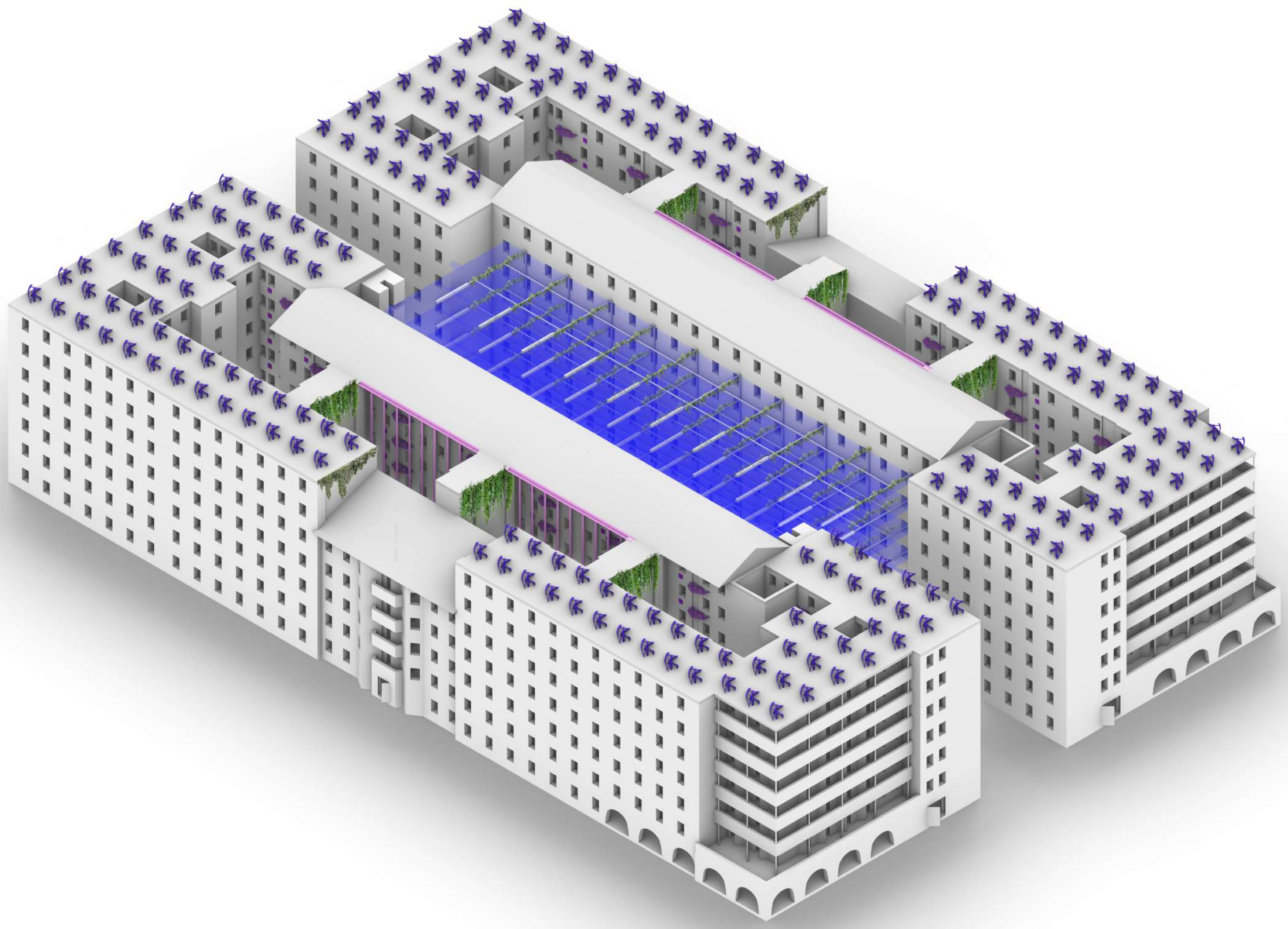
Casa de las Flores

Radio Wave Antenna

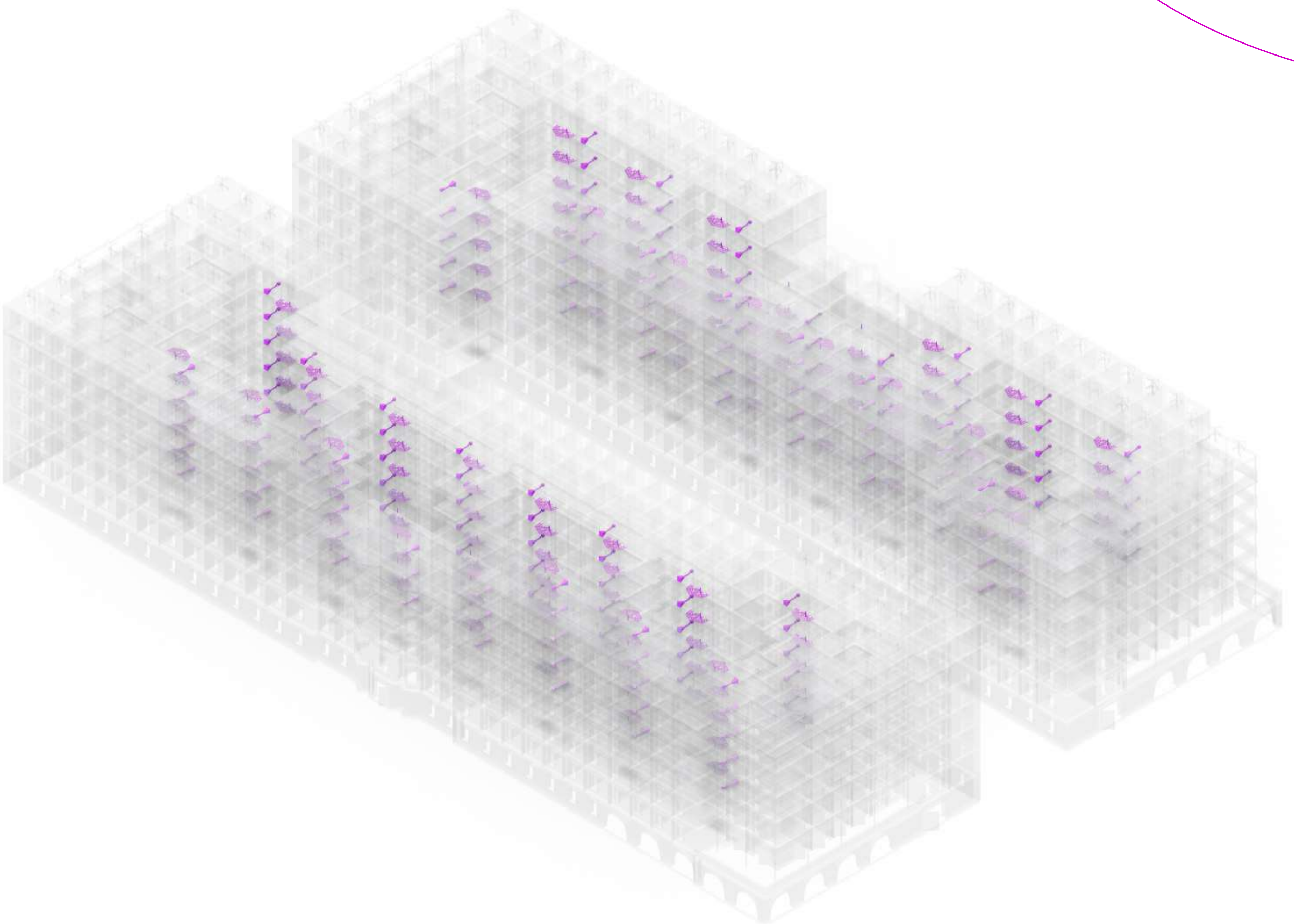
Wind Turbines

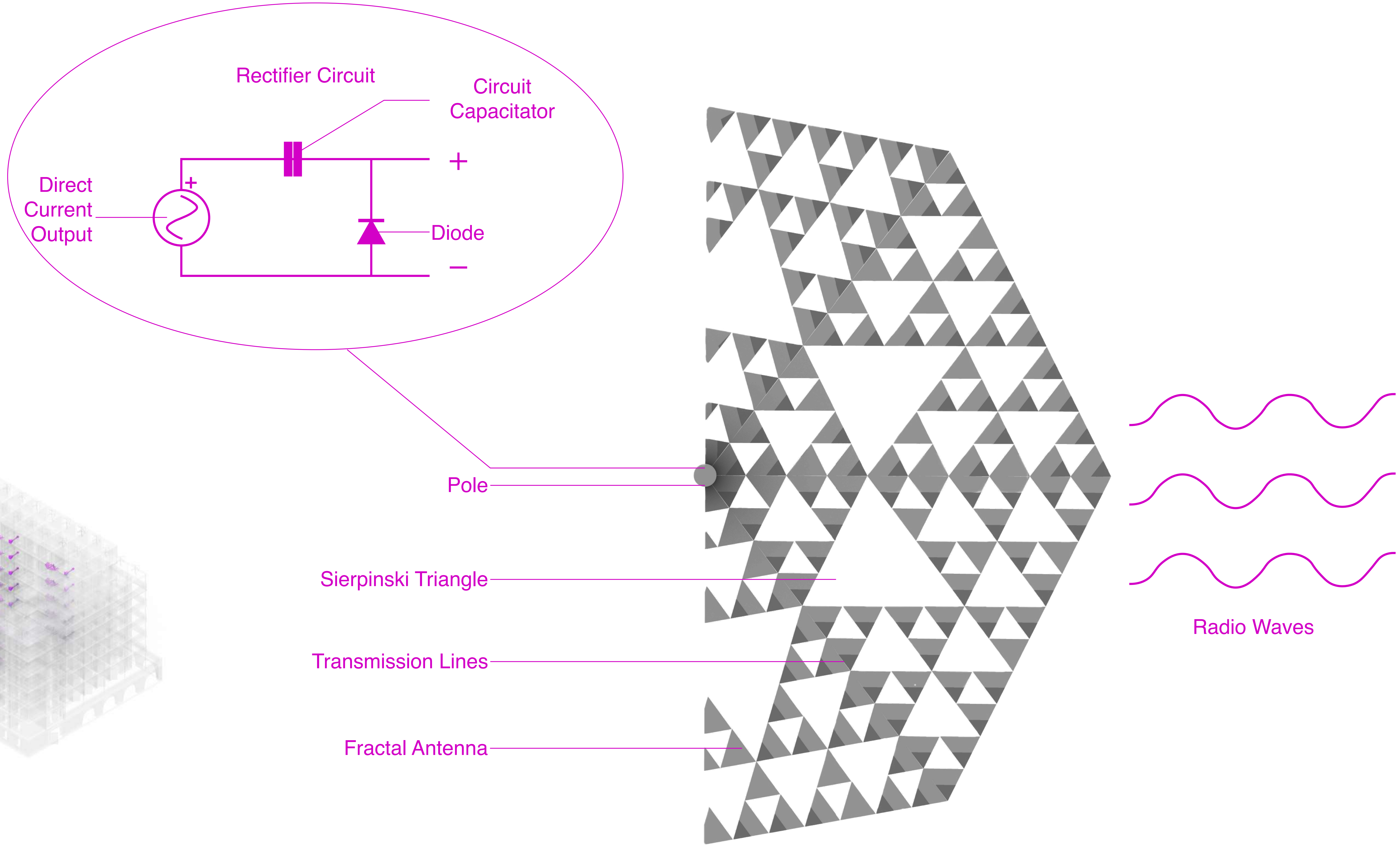
Hypogeum

Water Collection

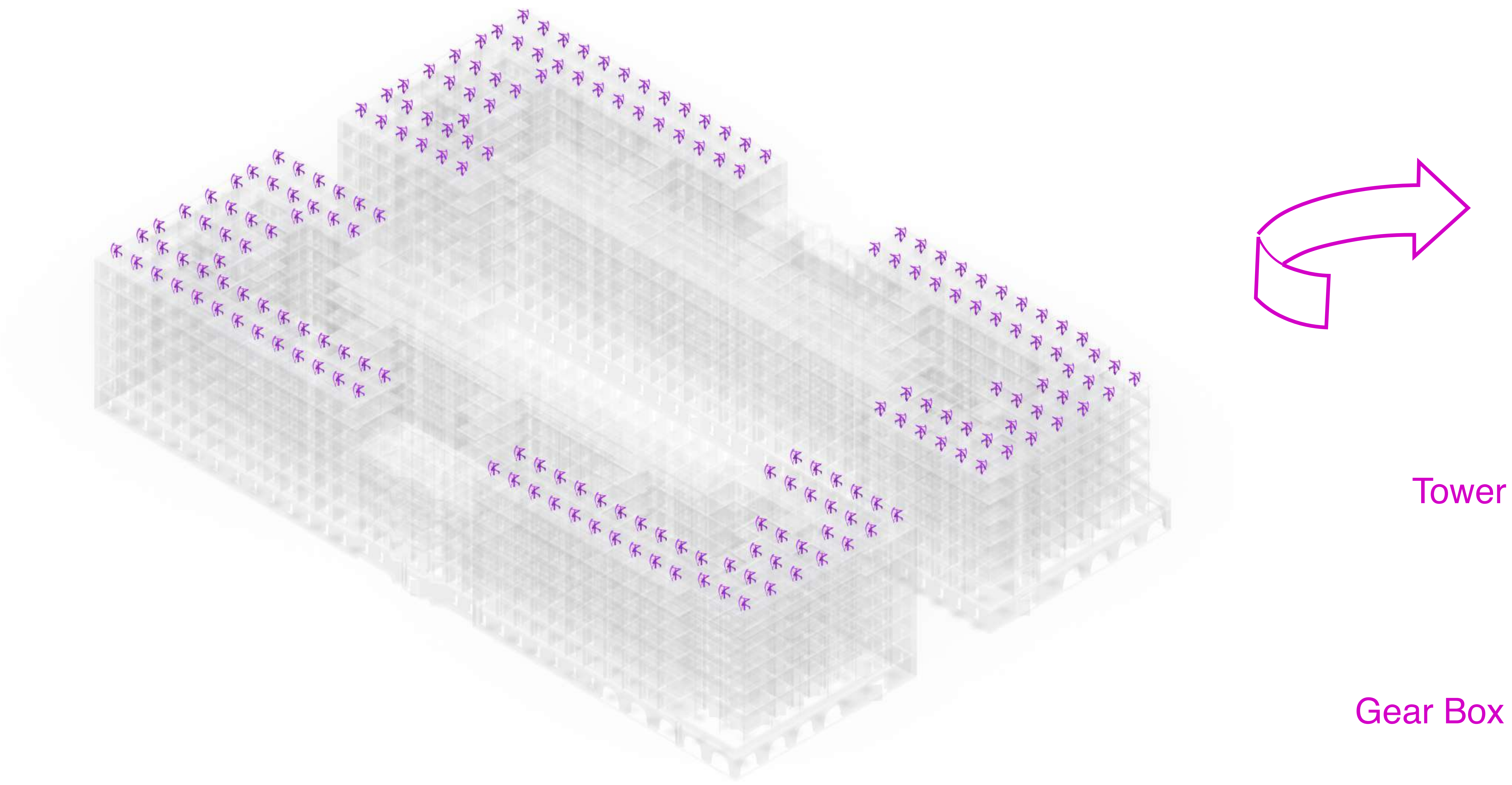


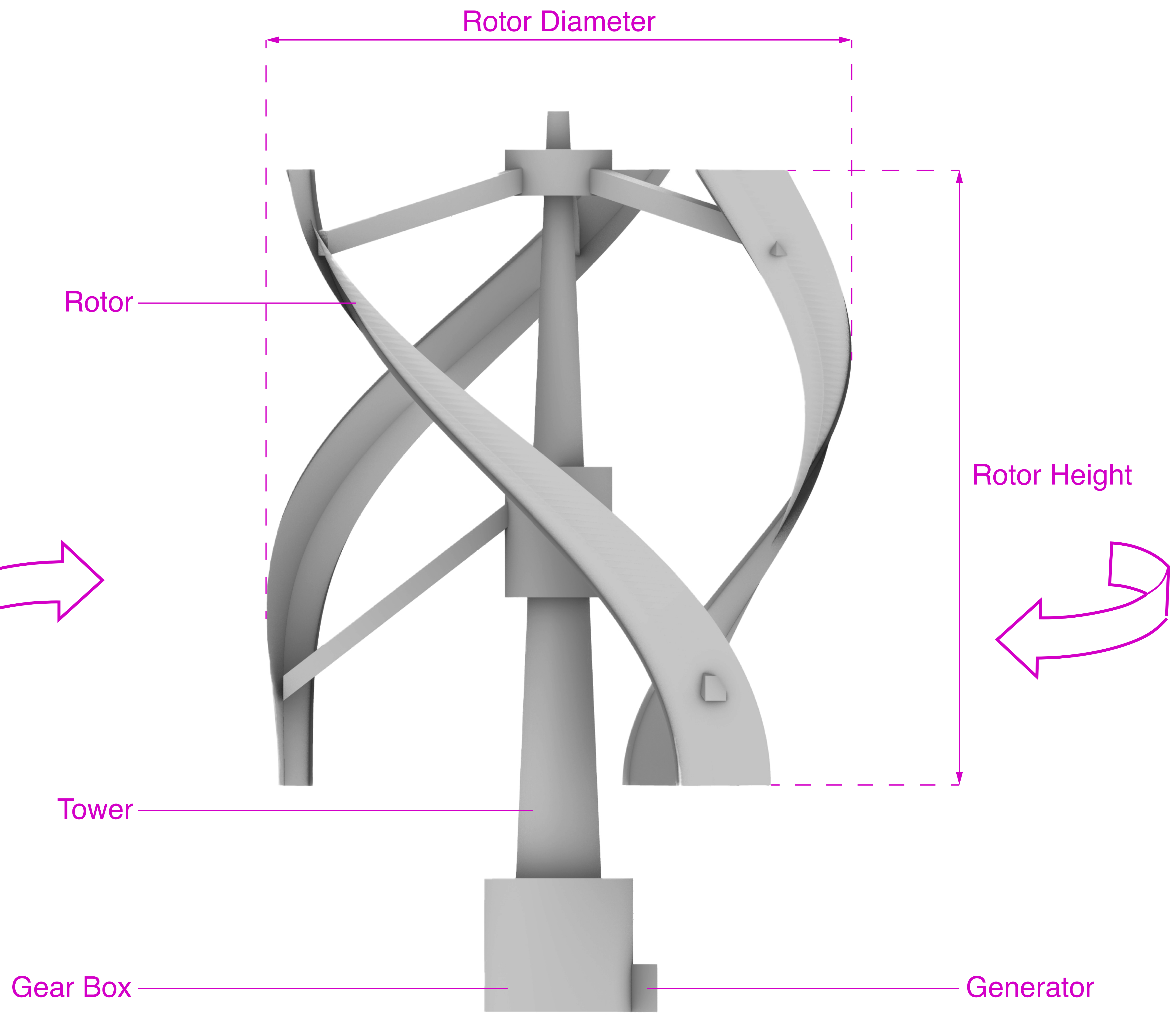
Radio Wave Antenna



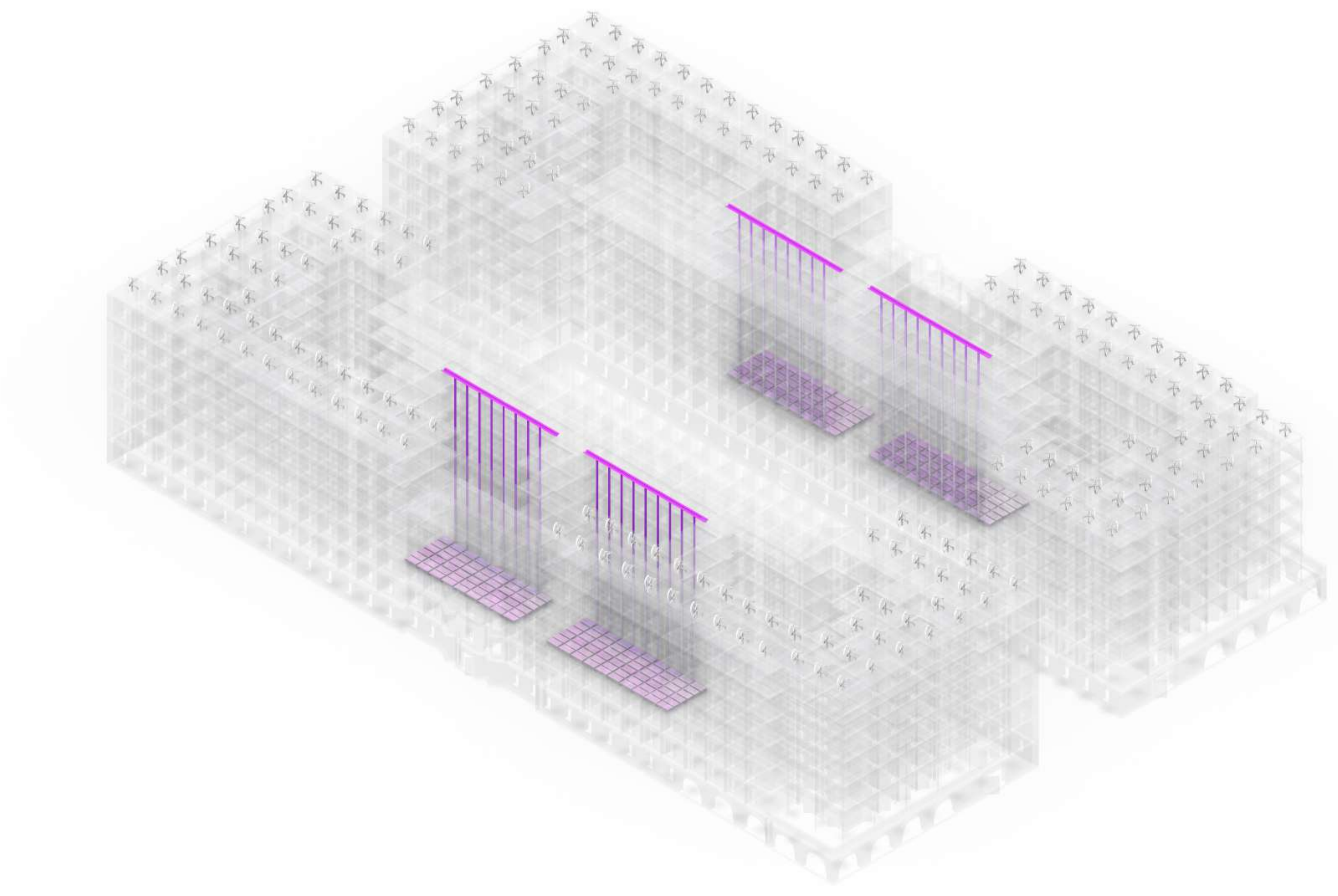


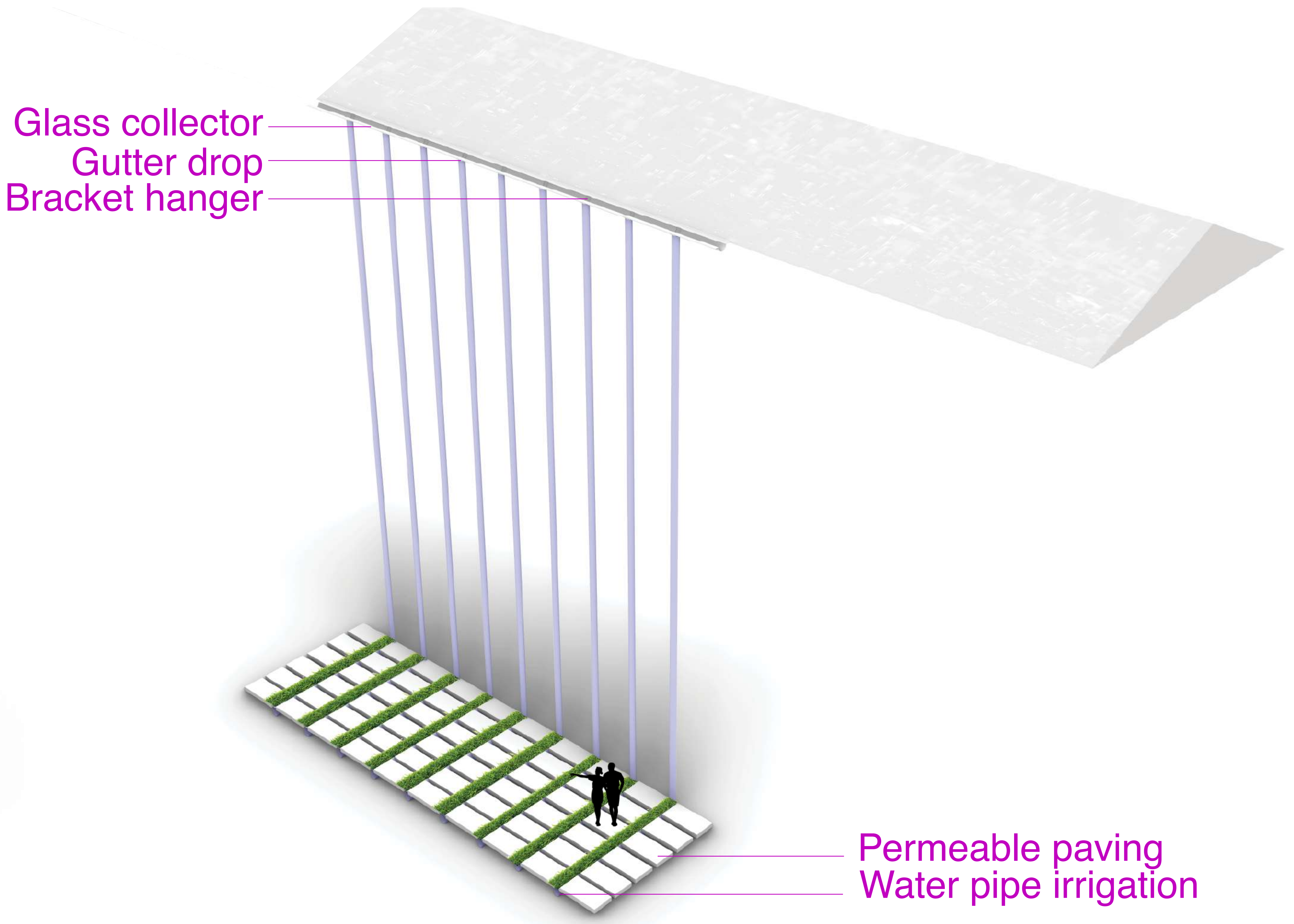
Wind Turbines

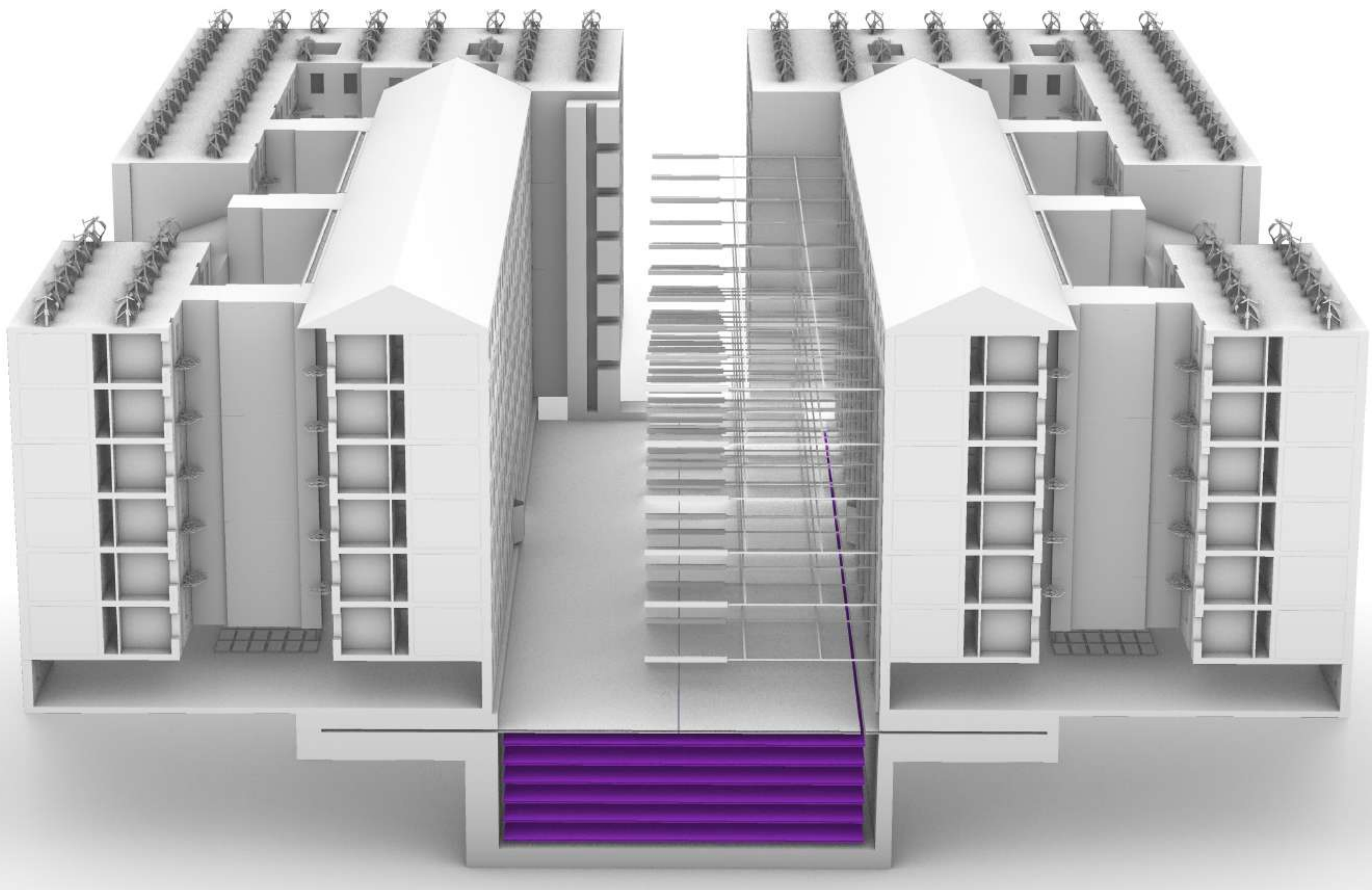


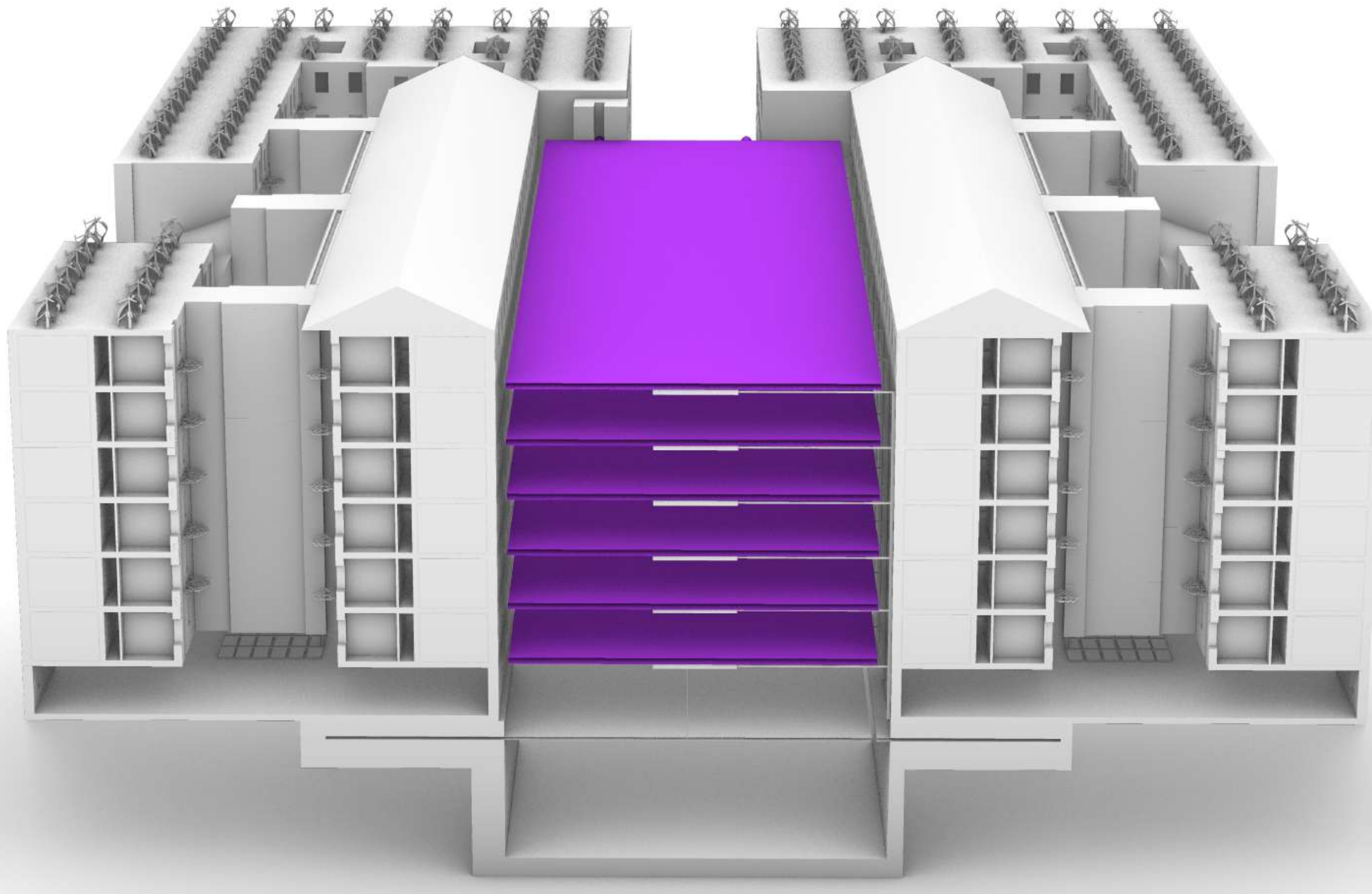


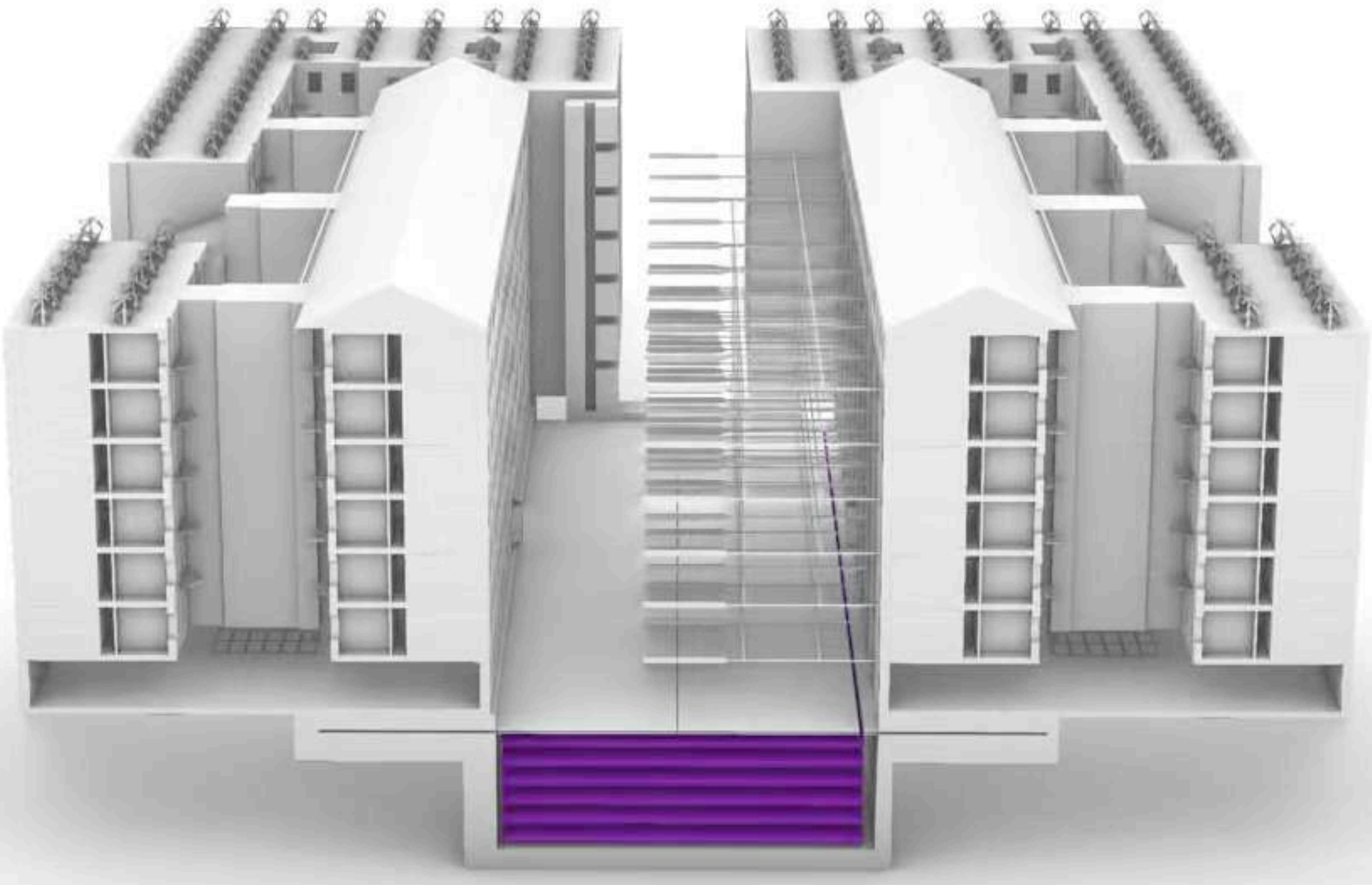
Water Collection



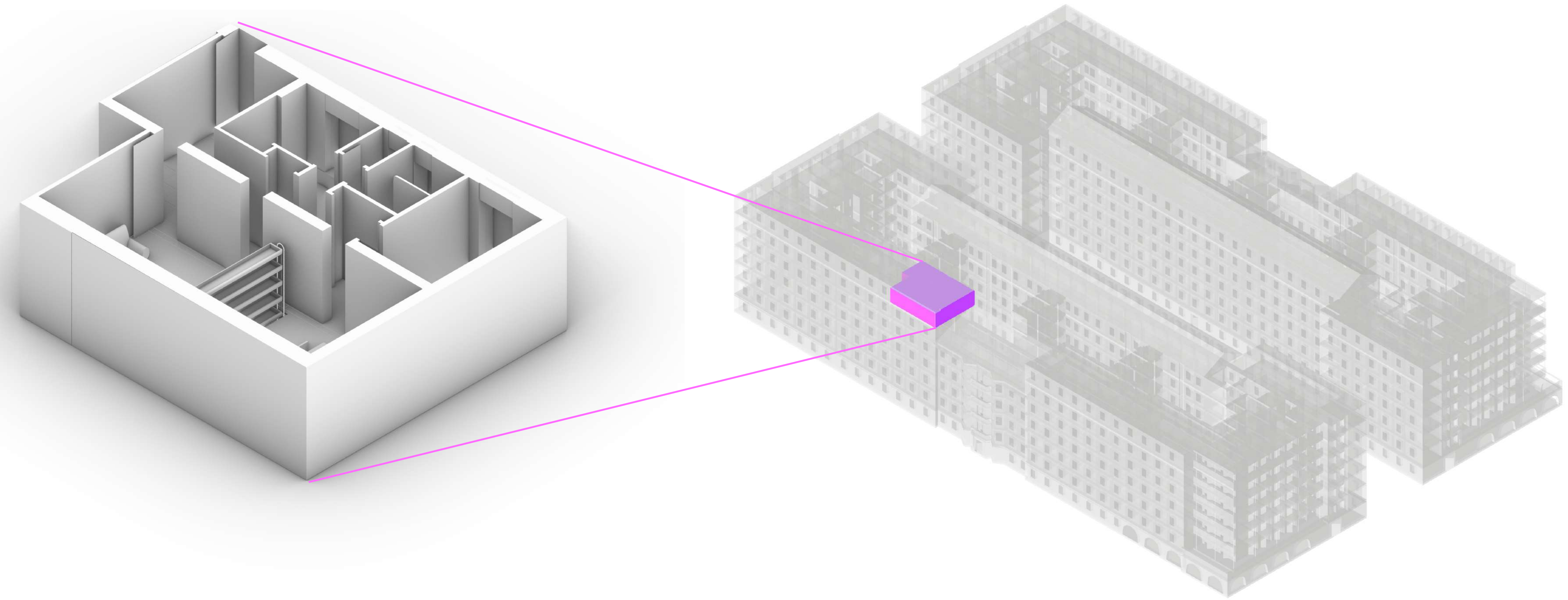








Unit 6C

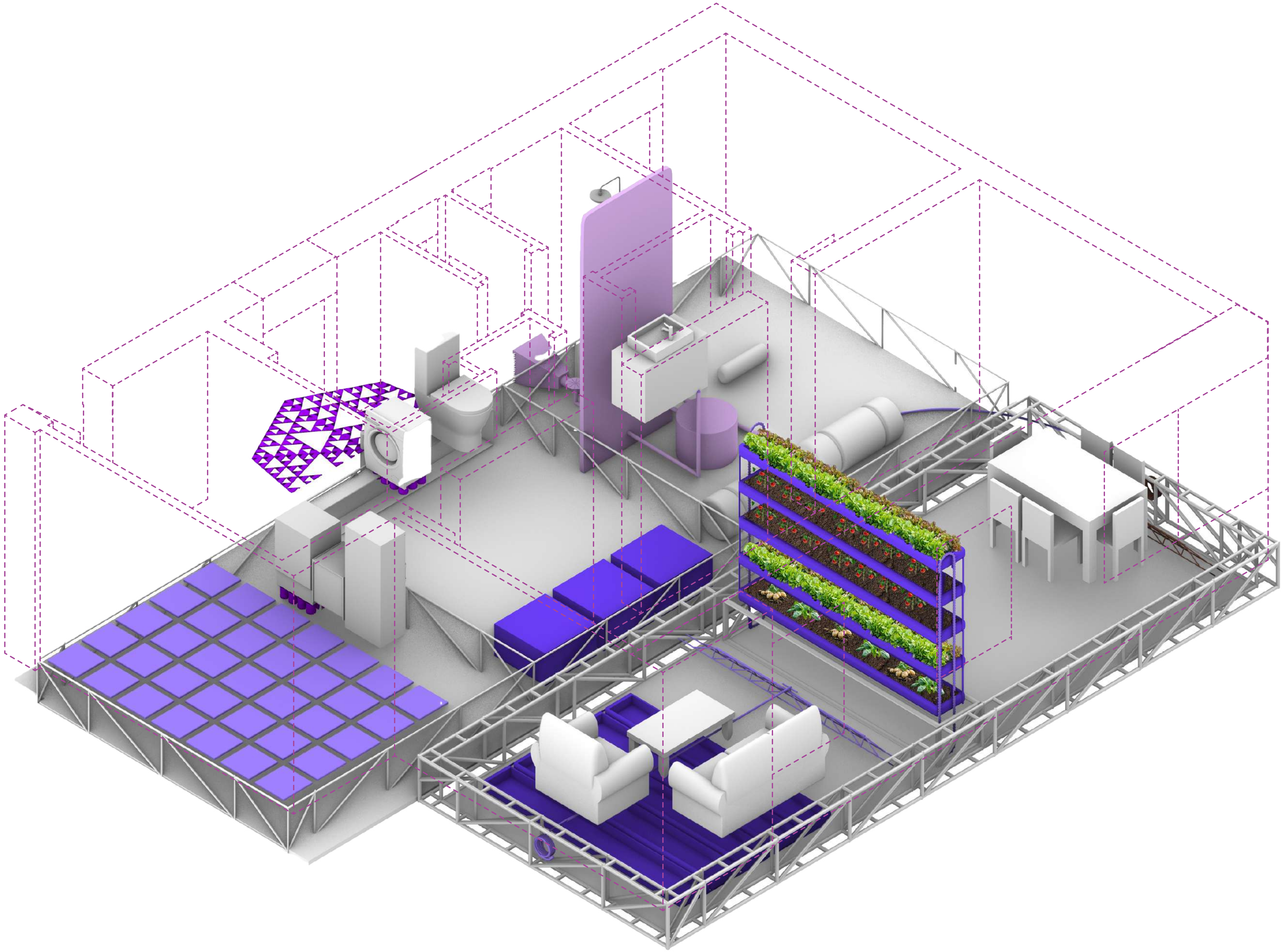


Unit 6C

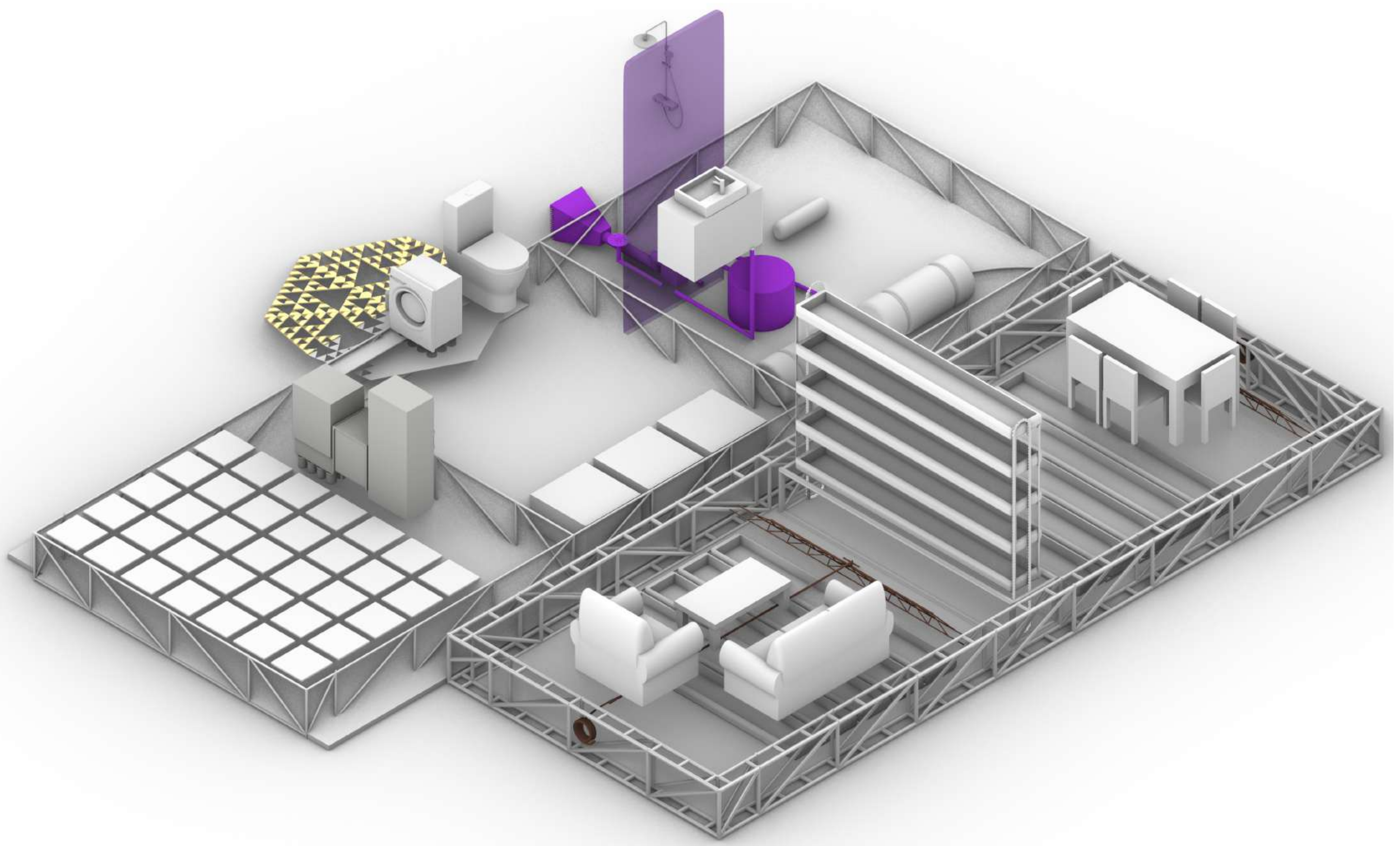
80m2

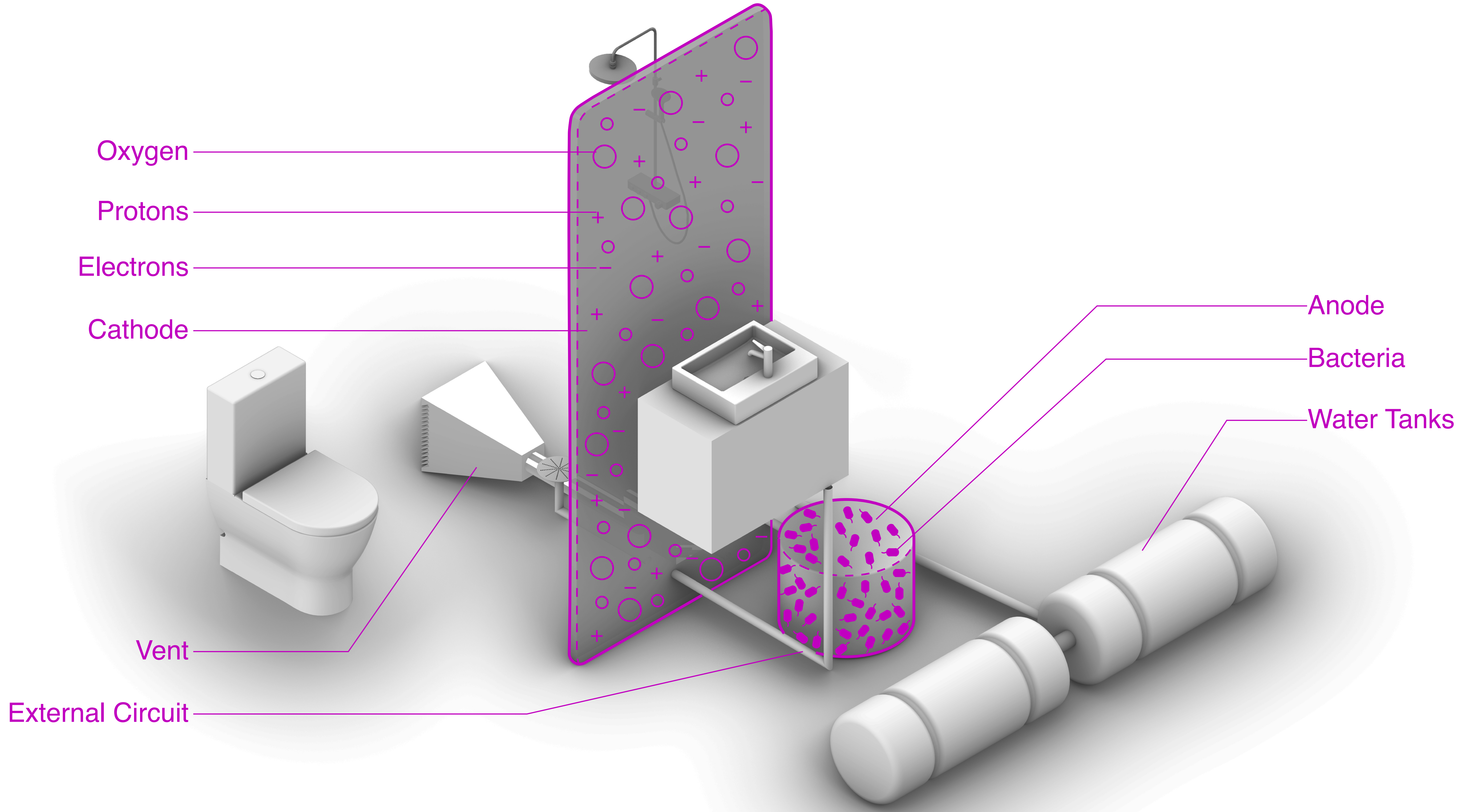
3 meter ceiling

288 units

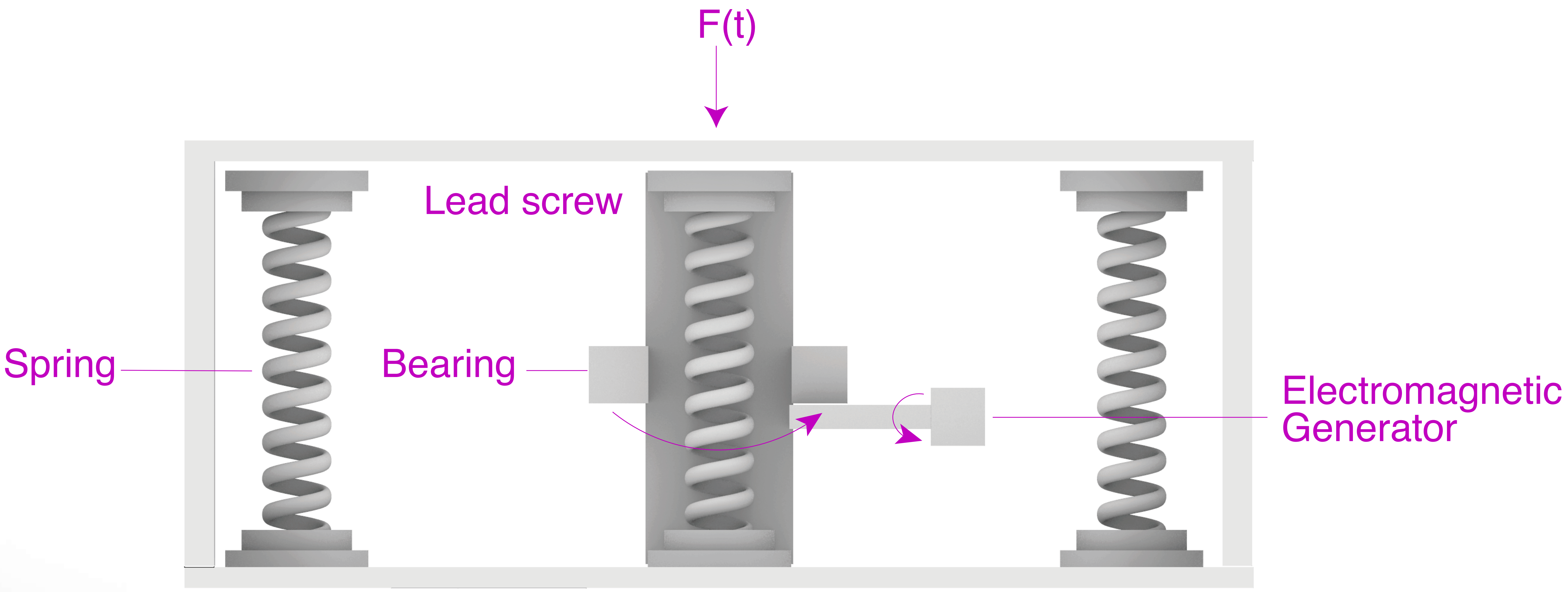


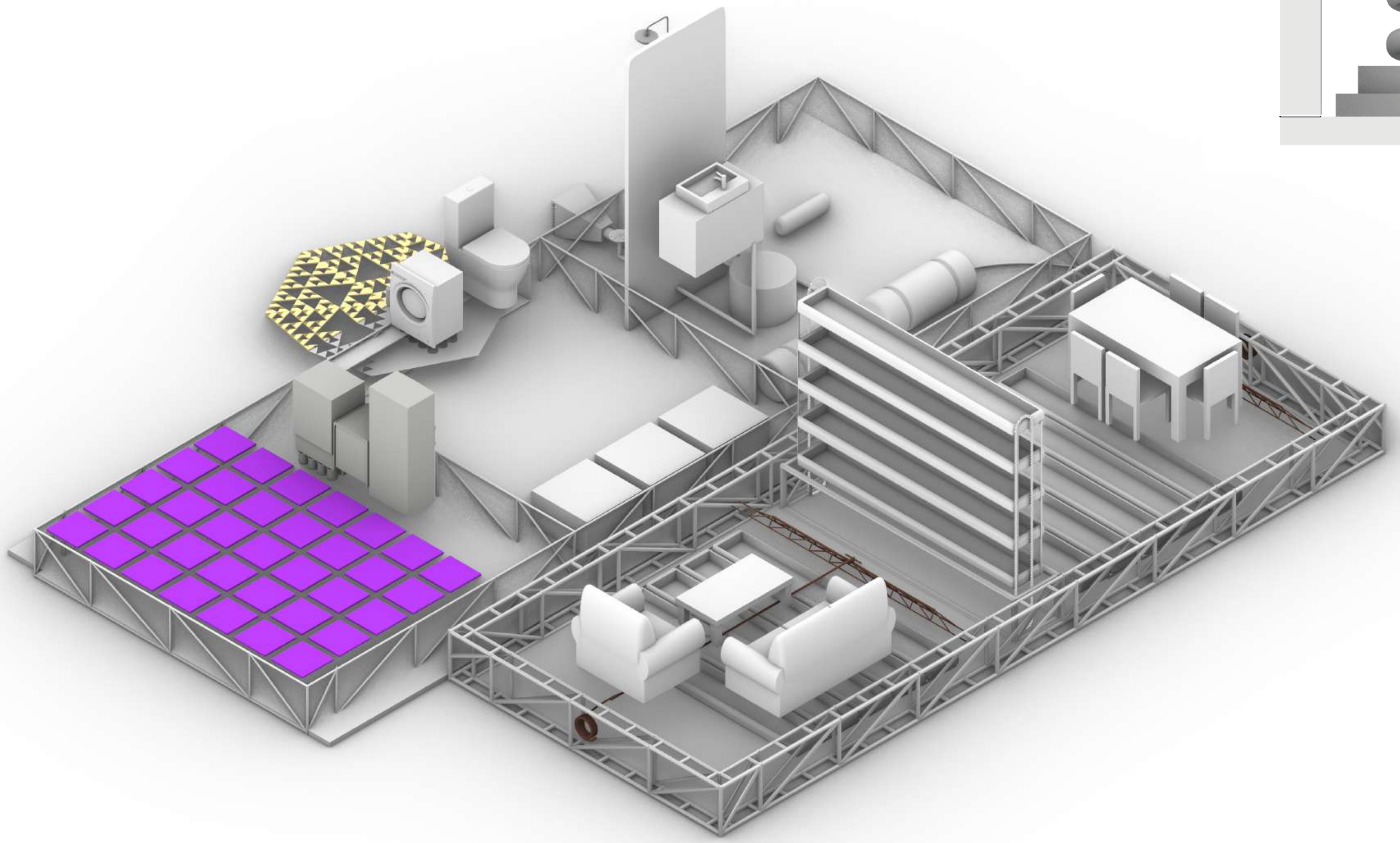
Microbial Fuel Cells



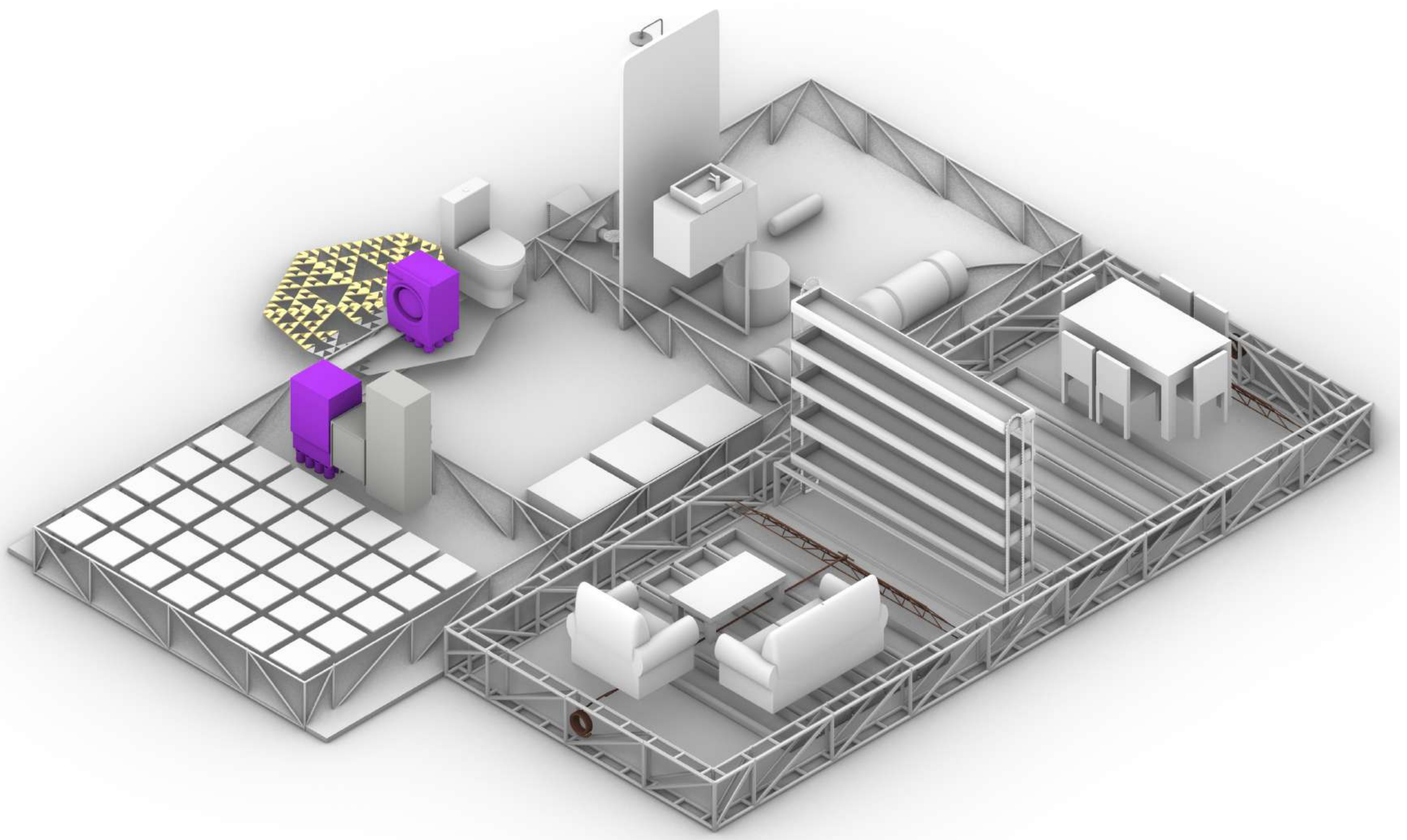


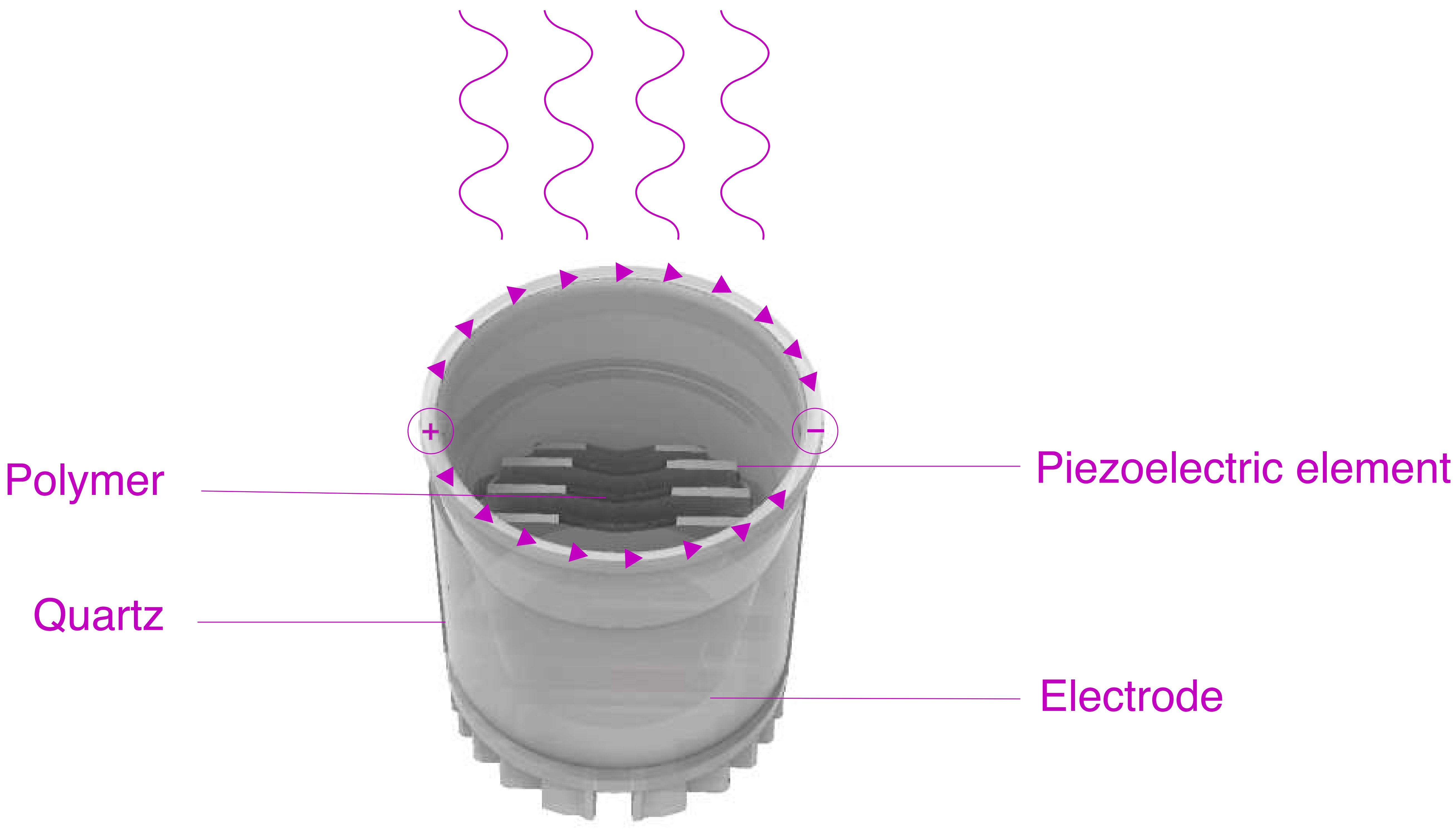
Kinetic Floor





Vibrational Harvester





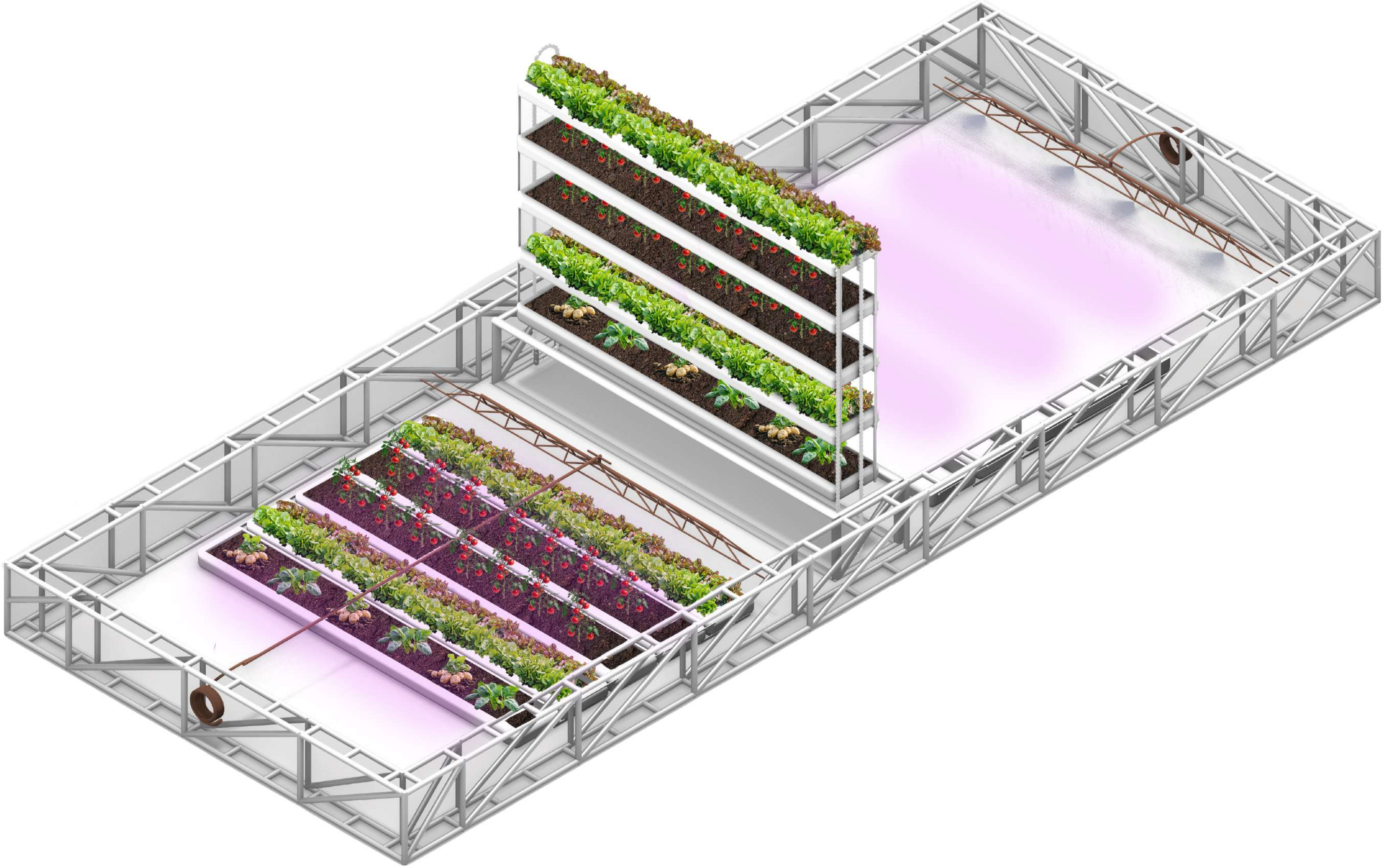
Hypogeum

2 Growing Platforms

2 Fertigation Pipes

32 LED grow Lights

8m²



Hypogeum

4 Tomato Plants

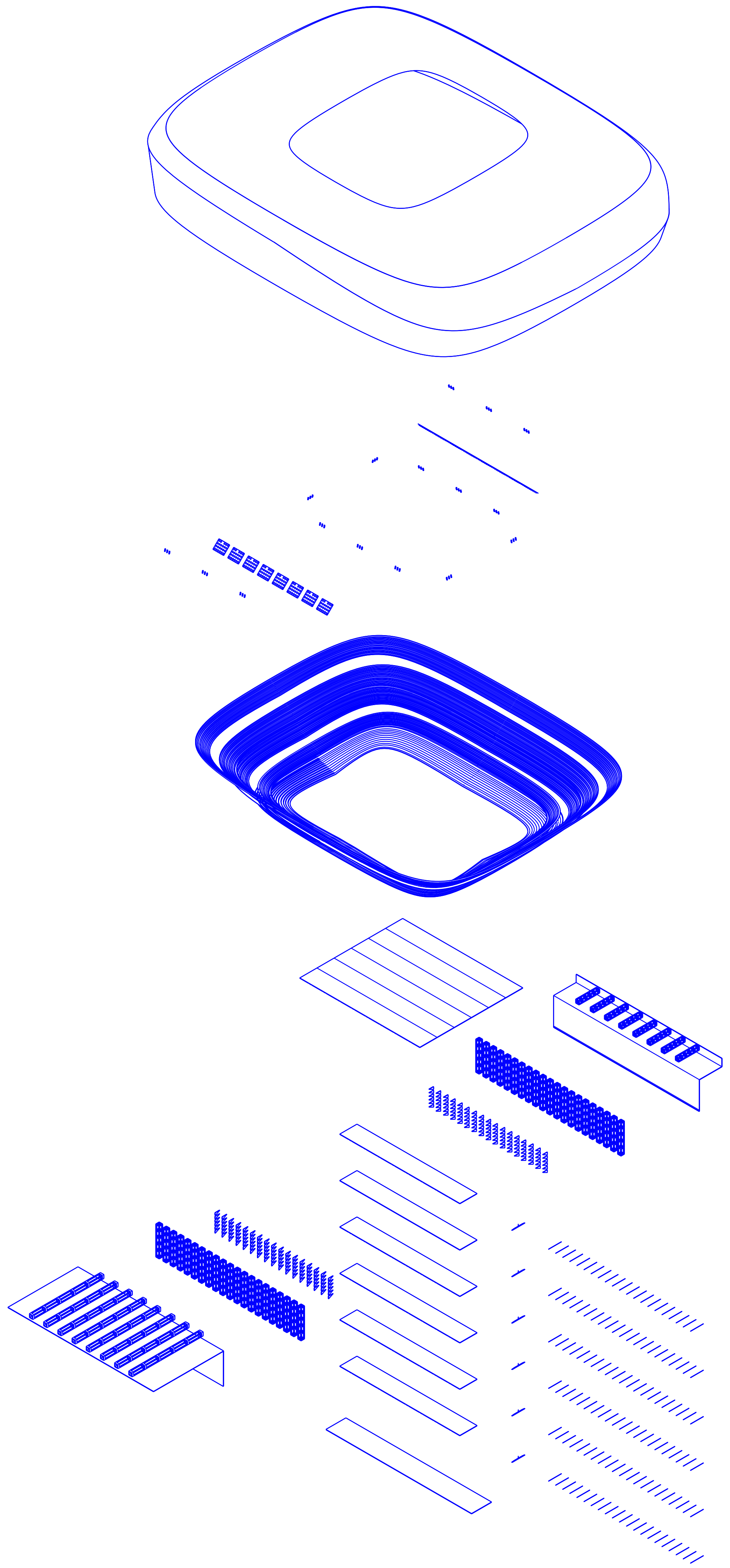
8 Pepper Plants

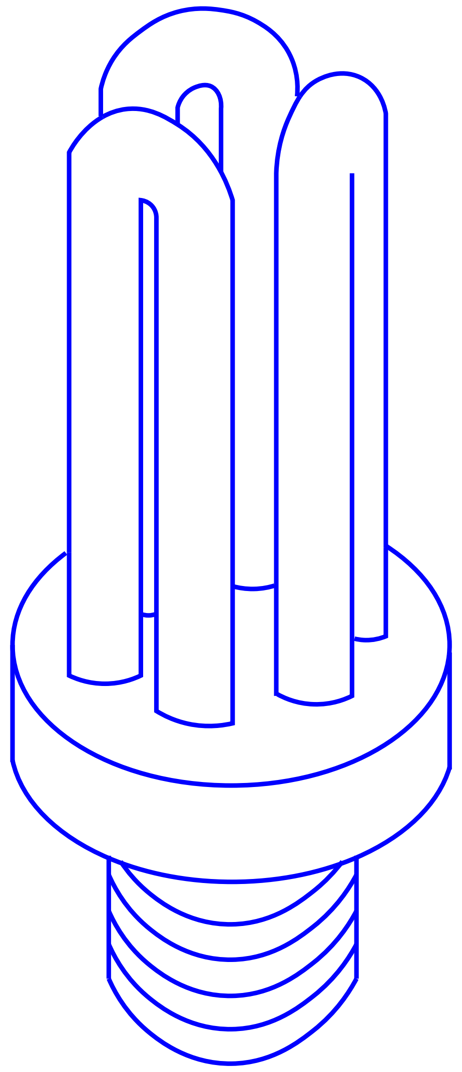
40 Potato Plants

20 Onion Plants

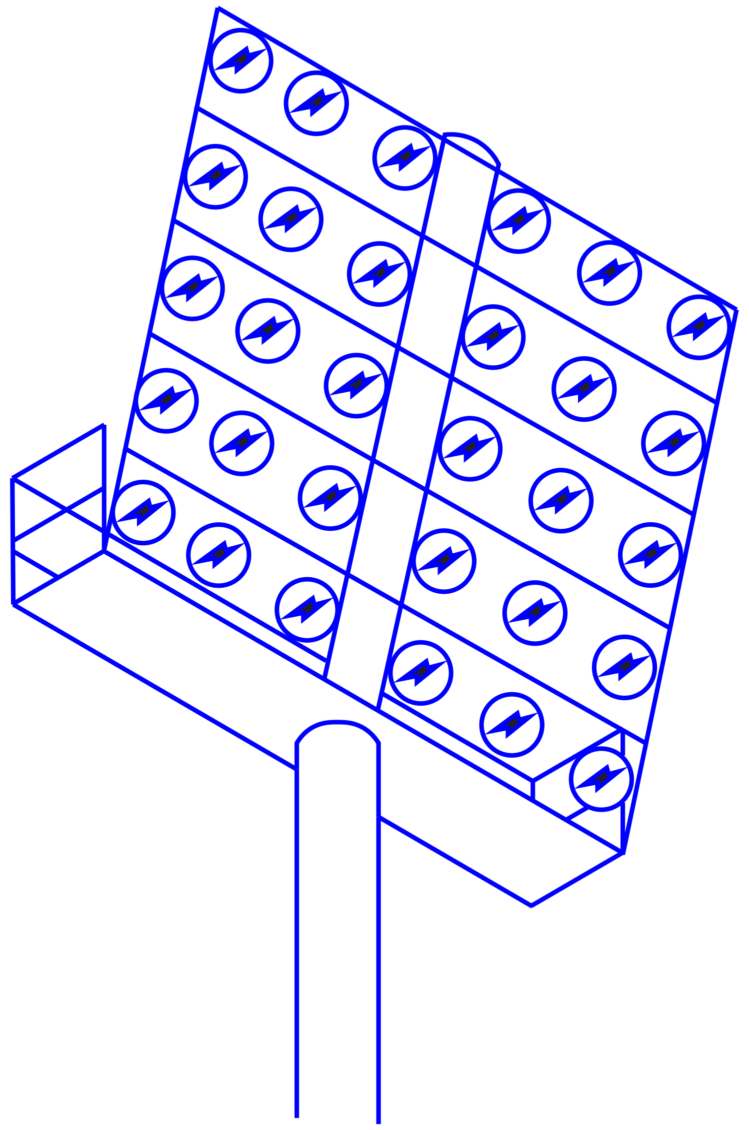
Feeds 2 people

Santiago Bernabéu

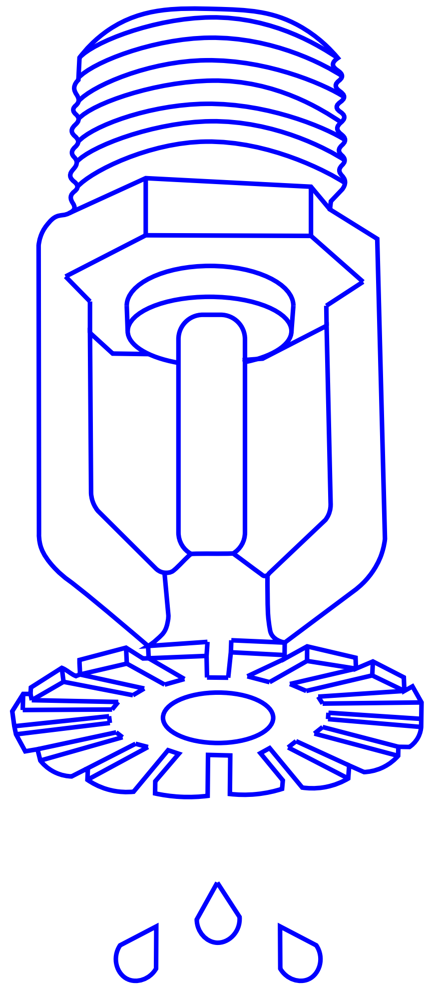




5,568 kWh

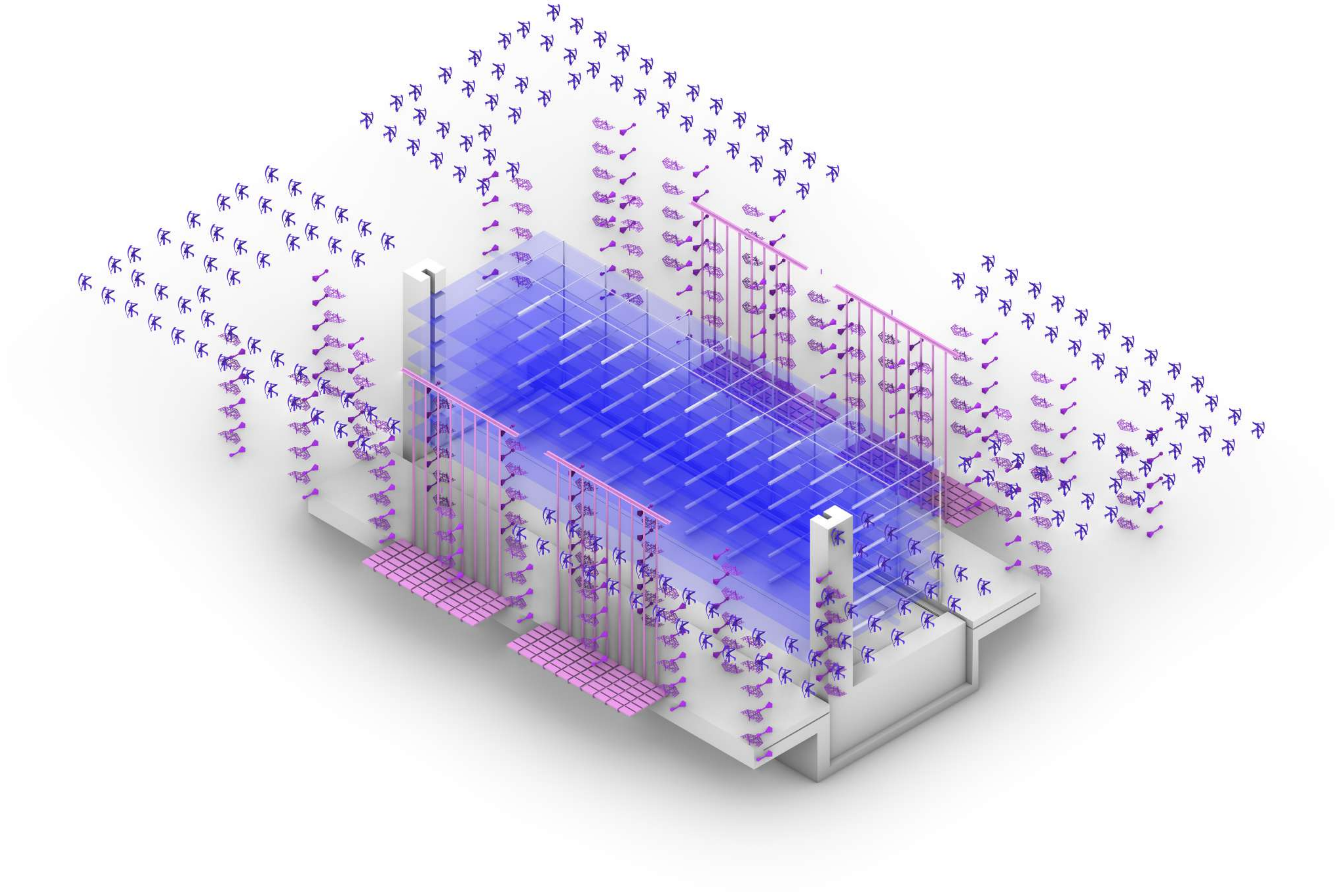


25,000 kWh



11,500 Liters

HomeGrown



Gutters collect:
20L/day

Antennas collect:
36.4kWh/day

Vibration Cells collect:
2.83 kWh / day

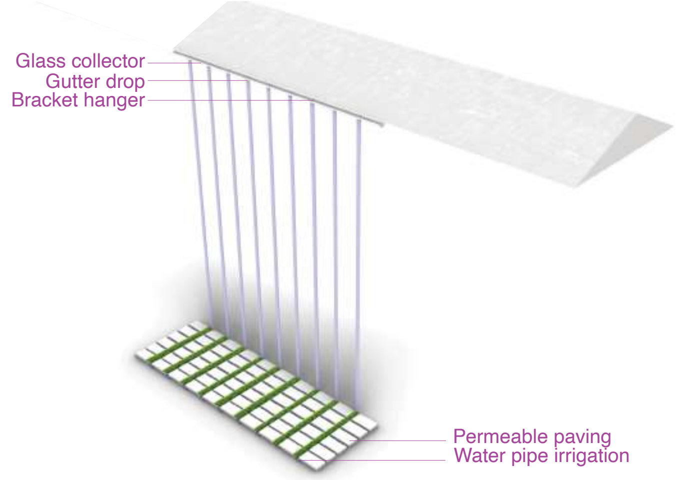
Turbines Collect:
13,900kWh/day

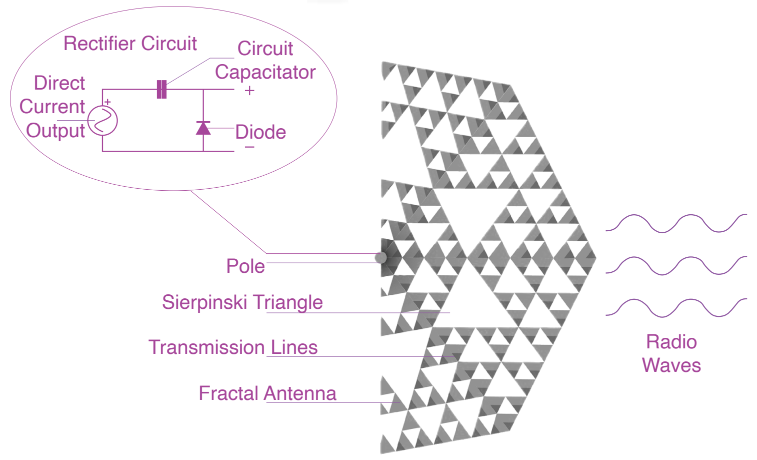
Microbial Cells collect:
38.4 kWh/day

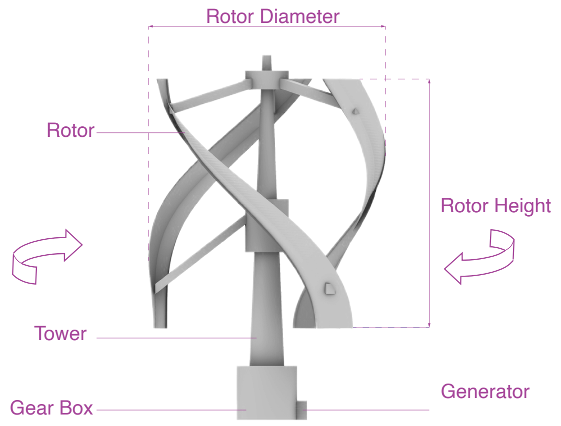
Kinetic Floors collect:
3.66 kWh / day

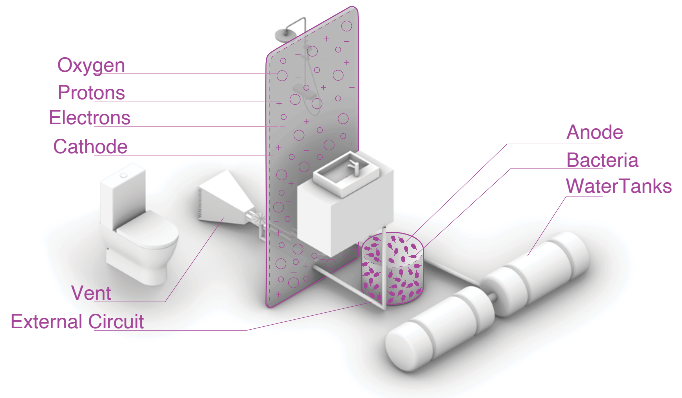
HOME GROWN

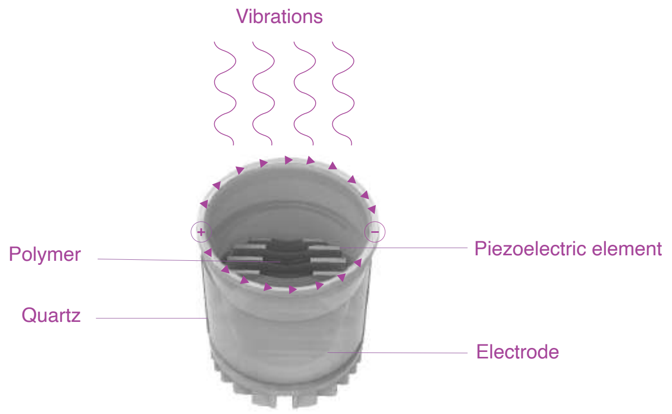
Nina Cohen, Isabel Sanchez, Kayla Vinh

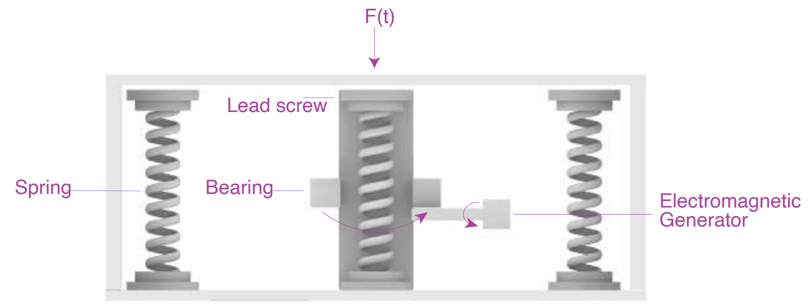


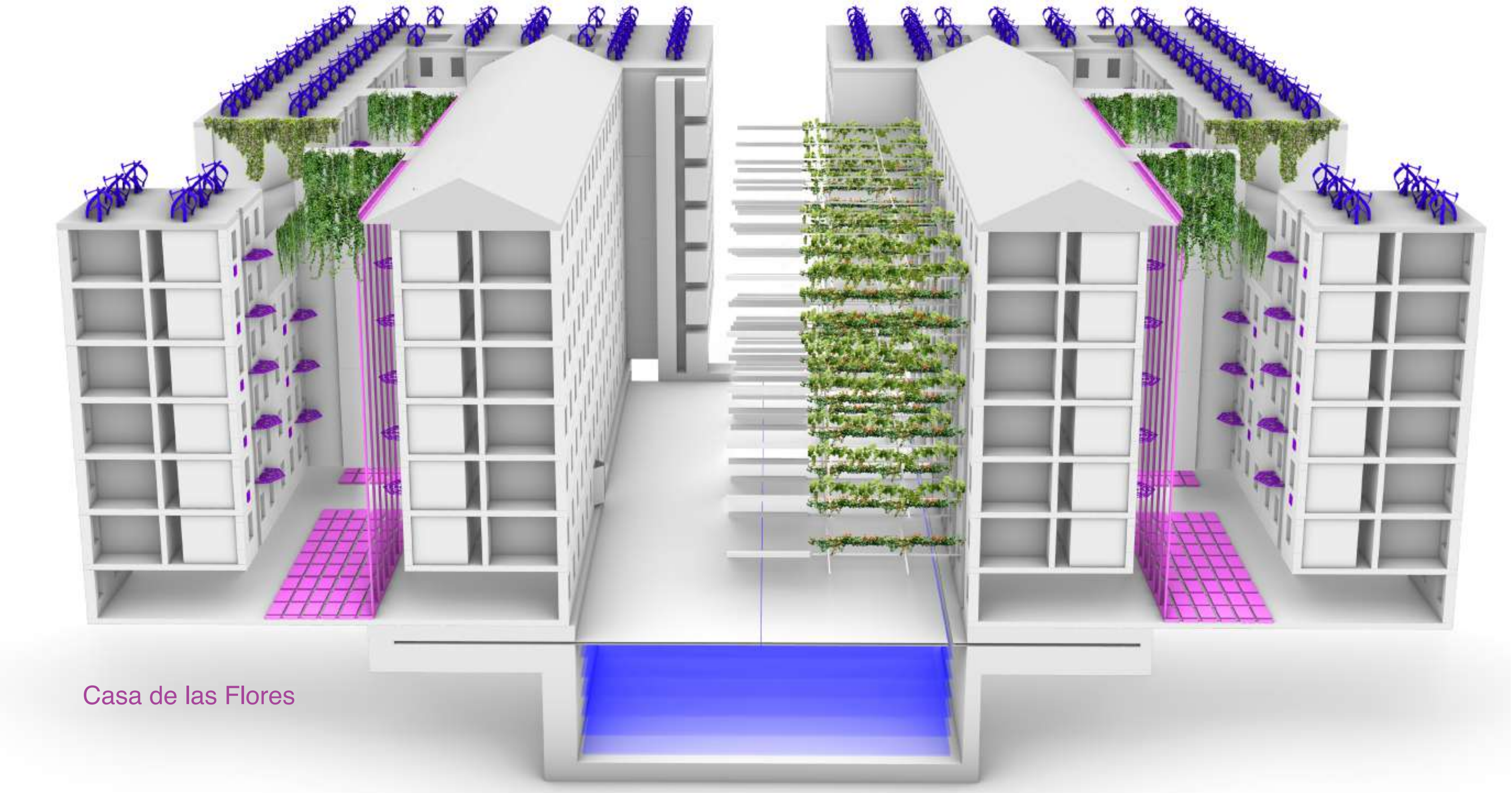




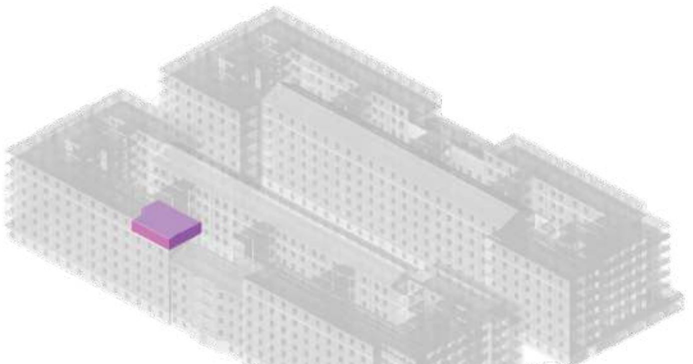




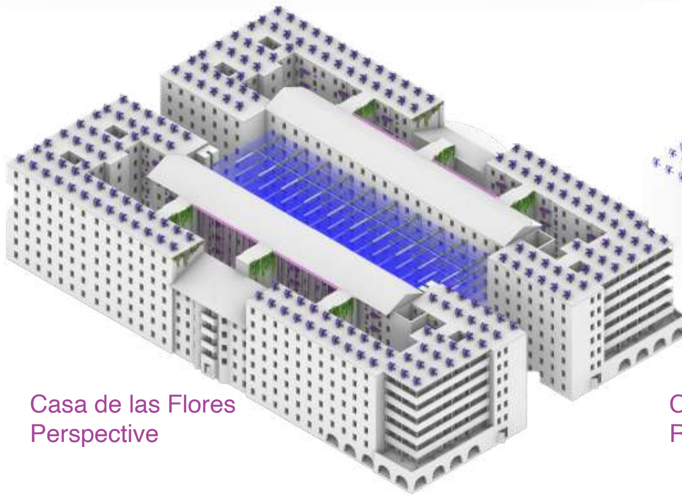




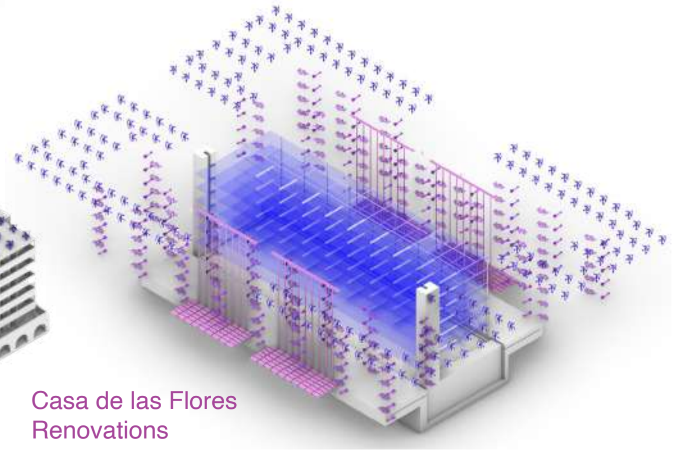
Casa de las Flores



Unit 6C



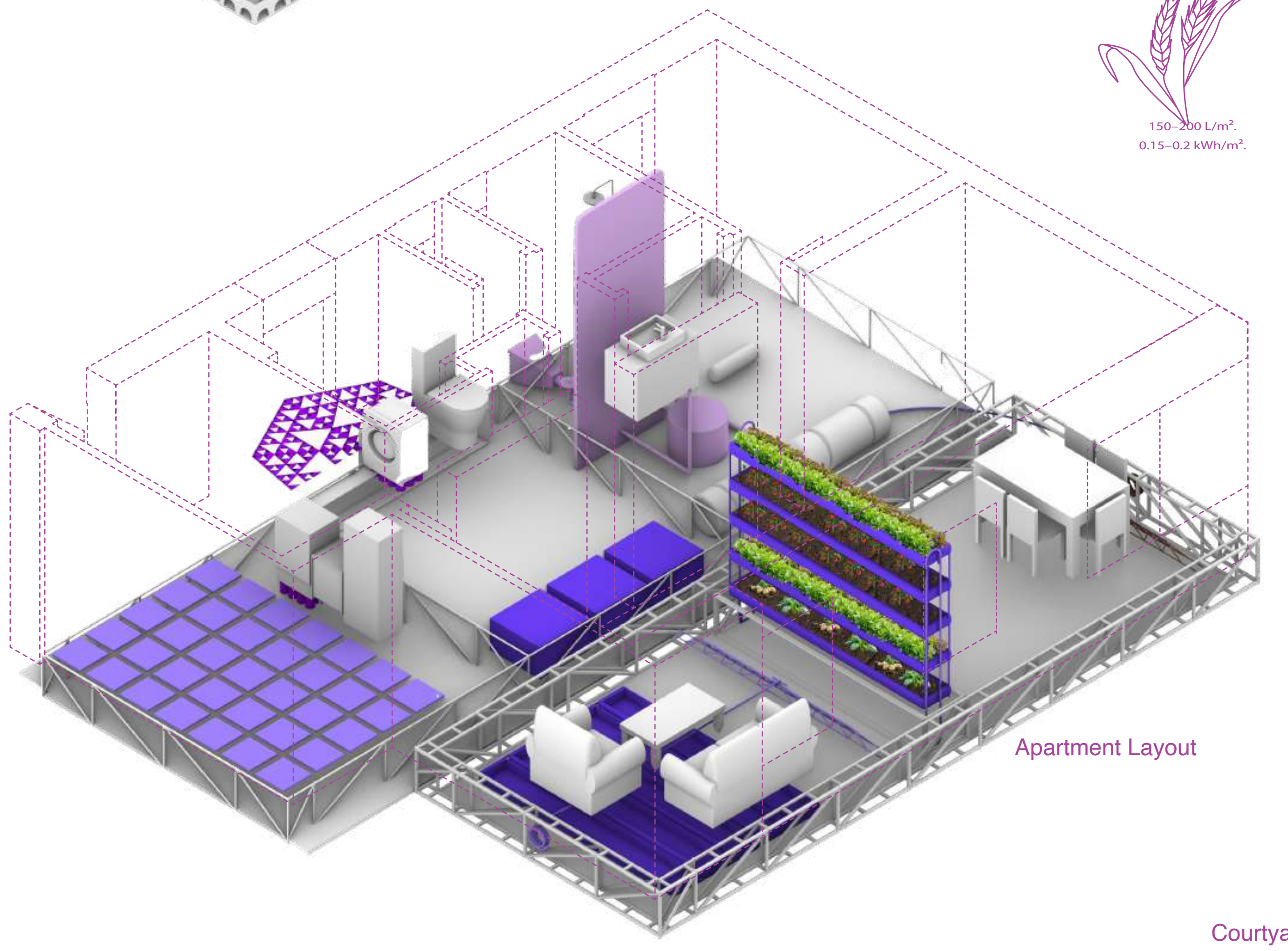
Casa de las Flores Perspective



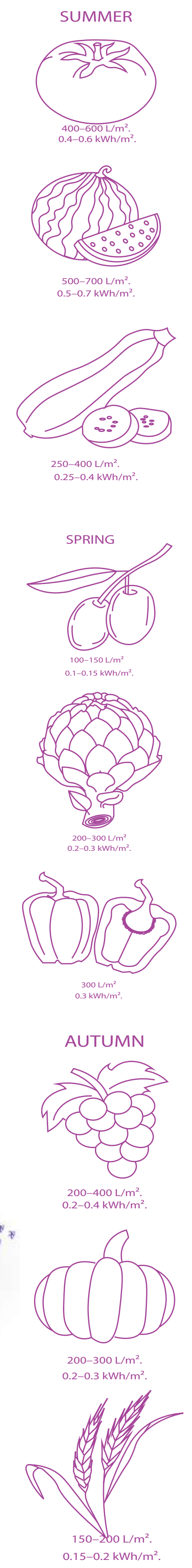
Casa de las Flores Renovations

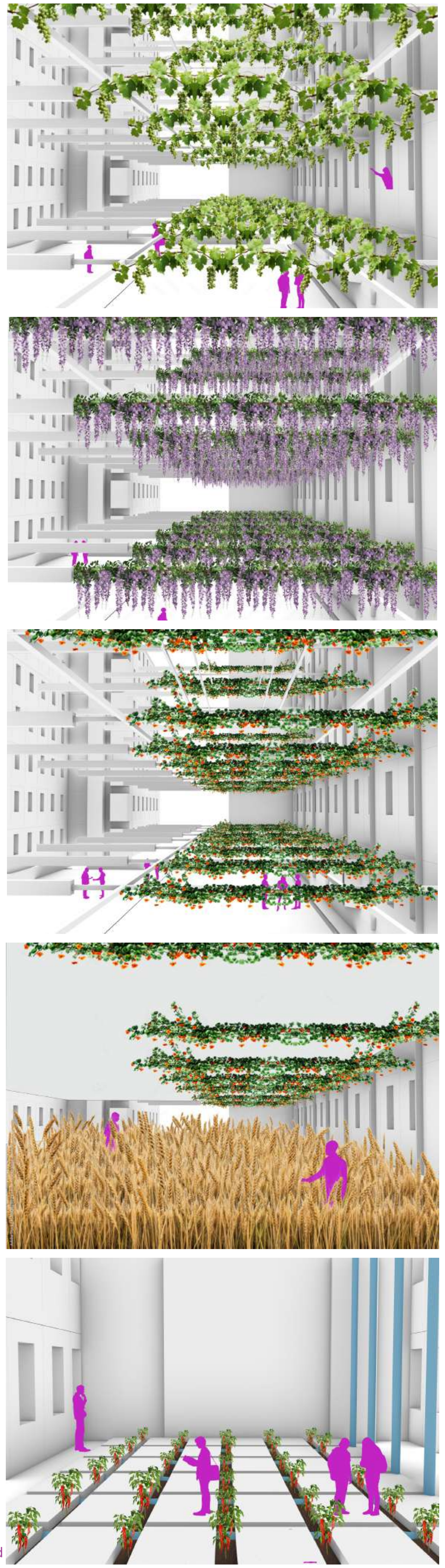
Living Room

Apartment Section



Apartment Layout





Courtyard

THANK YOU

The perfect home for you and your family in the case of sever famine. :)